P∞-E — The Normality of the Arithmetic State: the Entropic Route, the Pontryagin Index, and the Quantified Weld

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Every state splits into a part the filtration can see and a part that lives only at the horizon. The Hypothesis is that arithmetic keeps no secrets there.

State of the ledger (the referee's map — read this first). This paper was launched (June 11, 2026) as the execution of lemma (a) of the entropic tactic (P∞-D §7) and ended, two assault days later, as the record of three consecutive arcs. Method: GPD (Paper 35) — each section's closing "ledger" records the state of belief *at that strike*; superseded framings are flagged where they stand, never deleted. The arcs, with their fates:

Arc I (§§1–10, the entropic launch). The normality reformulation (E.2) → the Connes cocycle at the KMS half-boundary — *the ½ of the critical line is the ½ of modular theory* (§5) → the BC cocycle computed by sectors: diagonal and multiplicative *proven* boundary-regular, the obstruction localized to the additive×multiplicative interface (E.3) → the mixed kernel in closed form (Gauss sums × L'/L) and the exact circle (E.4) → the eigenfunctional decomposition and the entropy-growth exponent (E.5, E.6) → the 0–1 conjecture (E.7) → the assault: E.7 killed three ways, the Takesaki framing erratum'd to Arveson, the fake-curve No-Go (E.8), the three obligations (E.9), and the survivor E.7″ = Weil positivity = P∞-C pillar IV (§10).

Arc II (§§11–18, the index arc). Obligations (2),(3) unconditional in the Krein category (E.10, E.11); RH = Pontryagin index rigidity at $\kappa = 1$ (E.12) → the N_κ classical realization, RH $\iff$ $\kappa = 0$ (E.13) → index = Turing discrepancy, graded partial rigidity (E.14) → Krein-collision spectral flow, Honda–Tate = the collision locus (E.15) → the rank-one determinant: the open step as *one weighted uniform bound*, the weight = the archimedean envelope $e^{\pi|t|/4}$ (E.16) → the six-language dictionary (§16) → the Riemann–Siegel body: the weight is gauge, zero-protection is $O(1)$ in the Z-gauge (E.17) → one cone, two edges, the 1/8 seam (E.18).

Arc III (§§19–26, the weld arc). The CCM source read and welded (E.19) → the cliff-or-ladder self-attack (§20) → the exact matrix: exponentially-graded ladder, simple-even verified, r/ϵ measured (§21) → the quantified weld (E.20) and — landed on this pass — its erratum: the hidden hypothesis **(H0′)** (§22) → the Fuchs strike, (H3) reduced to the bridge (B2) (E.21) → the trace formula read at source (§24) → the radical route, (H2) decoupled (E.22) → the sector split, (B2) reduced to (B2b) (E.23). The sequence of ledgers:

$$\text{normality} \xrightarrow{\S10} \text{E.7}'' = \text{Weil positivity} \xrightarrow{\S\S11\text{–}12} \kappa\text{-rigidity} \xrightarrow{\S\S15,17\text{–}18} \text{one bound at the seam}$$
$$\xrightarrow{\S\S19\text{–}22} \text{(H1)+(H2)+(H3)} \xrightarrow{\S\S23\text{–}26} \text{(H1)+(B2b)+bookkeeping} \xrightarrow{\text{P}\infty\text{-F}} \text{(H0}')$$

Final form, up front: RH $\iff$ (H0′) — window Weil positivity surviving the limit, the hidden hypothesis of Theorem E.20 located by the P∞-F §14 audit (the erratum lands in §22) — through the corrected weld-amplifier. The fates of the other hypotheses: **(H1)** simple-even — **discharged unconditionally** (P∞-F §12, the Suzuki weld: his Theorem 5 = our E.11); **(H2)** — two proven lemmas plus certified bookkeeping (§25 here; E.22.3 = Lemma F.15 in referee form, P∞-F §16.1); **(H3)/(B2)** — sector-split (§26), every facet except (H0′) itself proven, certified, or welded (P∞-F §§7–16). The (H0′) anatomy — the three certified faces of ϵ_λ, the (H0′-tail) localization to the archimedean horizon, the horizon-form anatomy, the innovation chain, and the remainder law $R \approx Ce^{-0.342\lambda^2}$ — is P∞-F §§14–22.

Conventions of this draft. Historical flags mark superseded intermediate framings in place (the Takesaki reading of §§2–7 → the §10.1 erratum; E.6's exponents → E.6′, §10.4; E.7 → §10; §19.3(c)'s margin →

§§20–21; the §§23–26 ledgers → the final form above). Numbering: theorems run E.1–E.23 with lettered descendants (E.6′, E.7′′, E.21(a,b), E.22.1–3); §10's heading originally advertised "E.8–E.10" — corrected on this pass: E.10 opens §11. Anchors: the §15 block covers §§1–15; §§17–26 carry per-section anchors. Certificates: **sixteen probes**, fourteen in `empirical/p_infinity/` and two from the §10 assault (repo root; `empirical/`); the probe-by-section index closes the paper.

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1. The filtration and the two theorems that come for free

Let M be the type III_1 adele-class factor (BC at critical temperature; P∞-C §3) with the scaling action θ_t. For $\lambda > 1$ let

$$A_\lambda \;=\; W^*(\,\theta\text{-smeared scaling correspondences with budget} \le \lambda\,) \subseteq M,$$

the budget-λ observable algebra (the prolate-window corner's generated algebra). Then $\{A_\lambda\}$ is an **increasing net** and $\bigvee_\lambda A_\lambda$ is σ-weakly dense in M (the correspondences at all scales generate; routine [P]-sketch).

Let φ be the **arithmetic state** (induced by the prime side of the Weil pairing — the ζ-spectral data) and ψ the **free reference** (the pole/Gaussian side; N14's Gibbs state). Define the c-function

$$c(\lambda) \;:=\; S_{\mathrm{Araki}}(\varphi\|\psi\,;\,A_\lambda).$$

Theorem E.1 (the free part; [P], cited). (i) *Monotonicity:* $c(\lambda)$ is non-decreasing in λ — restriction to a subalgebra cannot increase Araki relative entropy (Uhlmann/Araki DPI; valid in arbitrary von Neumann algebras, type III included). (ii) *Martingale convergence:* **if φ, ψ are normal**, then $c(\lambda) \uparrow S(\varphi\|\psi; M)$ along any increasing net with dense union (continuity from below of relative entropy; Araki, Ohya–Petz, Hiai). □

So the c-theorem half of the entropic tactic is *free*: the monotone exists, with no operators built by hand.

2. The weld and the compression of the open step

[Historical flag — the Takesaki framing of this section (and of §7's tier upgrade) is corrected by the §10.1 erratum: at $\beta > 1$ the arithmetic state is normal and its Takesaki singular part vanishes identically; the residue functionals live in the Arveson θ-spectral decomposition, with the normal/singular dichotomy meaningful only in the critical limit. The compression's content survives in Arveson form; read E.2 accordingly.]

Stage identification [A]**.** CCM's stage-λ prolate positivity is the statement $c(\lambda) \ge 0$ in disguise: the N14 identity (archimedean Weil = relative entropy, Gibbs/Fenchel–Young) restricted to the budget-λ corner names the proven stage quantity entropically. (Extending N14 to the corners is work — lemma (a2) — but it is *finite-stage* work, the kind the program completes.)

The compression. Granting E.1(ii)'s hypothesis, the chain closes:

normality of $\varphi \implies c(\lambda) \uparrow S(\varphi\|\psi; M) \implies$ the limit functional is the exhausted one (no leakage) $\implies \mu_\lambda \downarrow 0 \implies$ RH.

Positivity already passes to weak limits; what RH needs — and all the program's no-go theorems say a finite stage cannot supply — is that the stages *exhaust* the limit object. Martingale convergence delivers exhaustion **iff the state has no part invisible to the filtration.** By **Takesaki's decomposition**, every state splits uniquely as

$$\varphi \;=\; \varphi_{\mathrm{normal}} \;+\; \varphi_{\mathrm{singular}},$$

with $\varphi_{\mathrm{singular}}$ vanishing on no finite stage's worth of σ-weak limits — supported, in the program's geometry, **on the horizon**.

Theorem-shape E.2 (the reformulation; forward direction conditional chain [A], converse [C]).

$$\boxed{\varphi_{\text{singular}} = 0 \ \text{ (the arithmetic state is normal)} \implies \text{RH.}}$$

The Riemann Hypothesis, on this route, is the assertion that **arithmetic stores no entropy on the horizon**: everything the primes know is visible to the scale filtration in the limit.

3. Why this is the right compression (consilience)

Every prior localization is this statement in another lens: the **Siegel shadow** (near-central structure invisible at bounded scale) is what a singular part looks like from inside; the **no-finite-witness theorems** (P∞-A 8.3a, P∞-B, the No-Go) say RH-certification cannot be normal-at-finite-stage — i.e., the question is *exactly* whether anything singular survives; the **logarithmic crawl** is the rate at which the normal part exhausts; Paper 32's **closure defect = beta source** vanishing is $\varphi_s = 0$ in the effective-action language; and the **second law** reading is literal: the singular part is entropy hidden from every observer — RH says the ledger balances because nothing is hidden.

4. The attack surface (next lemmas, in order)

- **(a1)** Density $\bigvee A_\lambda = M$ — routine; write it.
- **(a2)** Stage identification: extend N14's relative-entropy identity to the budget-λ corners (finite-stage work; the CCM positivity is the target's known face).
- **(a3) The question:** compute or bound $\varphi_{\text{singular}}$. Concrete handles: (i) Takesaki's criterion — φ is normal iff σ-weakly continuous iff completely additive; test against the explicit prime-side functional (the $\Lambda(n)$ ledger); (ii) the singular part of a weight on a III_1 factor pairs with the flow of weights — the program's trace-scaling identifications (P∞-C §3) give the computational frame; (iii) **falsifier**: if a nonzero φ_s is *forced* by the Euler product's divergence structure, this route dies and the obstruction is named — either outcome is decisive for the route.

Honest status. E.1 is cited theory; the weld (a2) and the equivalence's converse are open; E.2's forward chain is conditional on the identifications. Nothing here proves RH. The deliverable is the compression: the open step is now *one question about one state on one factor* — normality — sitting in a toolbox (Takesaki decomposition, flow of weights, martingale convergence) that the program has not yet exhausted and that was built for precisely this type of object.

5. (a1) discharged; (a3) sharpened to the Connes cocycle at the KMS half-boundary

(a1) Density [P-sketch]. The smeared scaling correspondences over *all* budgets generate the crossed product by construction (the θ-smeared multiplication operators with unrestricted test support are the defining generators of M); each generator is a σ-weak limit of its budget-λ truncations (normality of the defining representation), so $\bigvee_\lambda A_\lambda \supseteq$ a generating set, hence $= M$. □

(a3) sharpened. Normality is representation-relative: φ is tautologically normal in its own GNS representation. The genuine question is the **relative position** of the arithmetic state and the free reference: Sakai–Connes absolute continuity $\varphi \ll \psi$, whose carrier is the **Connes cocycle derivative** $[D\varphi : D\psi]_t$ — existing for all real t by general theory, with the content concentrated in its **analytic continuation across the KMS strip**. And here the program's oldest constant surfaces in its native habitat: the strip's boundary is $\operatorname{Im} t = -\frac{1}{2}$ — the Tomita–Takesaki $\Delta^{1/2}$ point. **The $\frac{1}{2}$ of the critical line is the $\frac{1}{2}$ of modular theory.** The reformulation, sharpened:

$$\boxed{\text{RH} \iff \text{the cocycle } [D\varphi : D\psi]_t \text{ extends regularly to the boundary } \operatorname{Im} t = -\tfrac{1}{2} \ \ (\varphi \in \operatorname{dom} \Delta_\psi^{1/2} \text{ relatively}).}$$

(Tier: the $\iff$ as stated is [A] — the forward weld through E.2's chain; the precise equivalence is the write-out target.) This is consilient with everything modular the program owns: FE = time-reflection (P∞

§5), the modular adjudication and Prop 8.3a (P∞-A §8 — finite modular witnesses provably impossible: exactly a cocycle-boundary statement), and N14's relative entropy (the cocycle's $-i$-derivative at $t = 0$).

The next computation (concrete, finite, queued). On the BC factor the modular theory is explicit and the partition function of the modular flow is $\zeta(\beta)$ itself: the cocycle $[D\varphi : D\psi]_t$ is expressible in ζ-data, and its continuation to $\operatorname{Im} t = -\frac{1}{2}$ probes precisely the critical-line boundary. Compute the cocycle explicitly on the BC system; its boundary behavior at $-\frac{i}{2}$ is the route's first genuinely *calculable* face of the open step — and the falsifier stands: a forced singularity at the boundary, derivable from the Euler product alone, would kill the route with the obstruction named.

6. The BC cocycle, computed by sectors: two regularity exclusions and the localization (Prop E.3)

On the BC GNS space $\ell^2(\mathbb{N}^\times)$, $H|n\rangle = \log n\,|n\rangle$, $Z(\beta) = \zeta(\beta)$; isometries $\mu_m|n\rangle = |mn\rangle$; Gibbs reference $\psi_\beta = \zeta(\beta)^{-1}\sum_n n^{-\beta}\langle n|\cdot|n\rangle$; arithmetic (von Mangoldt) state $\varphi_\Lambda = (-\zeta'/\zeta(\beta))^{-1}\sum_n \Lambda(n)n^{-\beta}\langle n|\cdot|n\rangle$.

(i) Diagonal sector: regular at the boundary. Both states are diagonal; the cocycle is the diagonal unitary

$$[D\varphi_\Lambda : D\psi_\beta]_t\big|_{\text{diag}} = \left(\tfrac{\Lambda(n)\,\zeta(\beta)}{-\zeta'/\zeta(\beta)}\right)^{it},$$

whose continuation to $t = -\frac{i}{2}$ has symbol $\left(\Lambda(n)\,c_\beta\right)^{1/2} \le (c_\beta \log n)^{1/2}$ — **logarithmic growth only**: relatively bounded by every H-power; no singular part. *The diagonal (density/ledger) sector is absolutely continuous.* □

(ii) Multiplicative sector: entire. The flowed two-point function of an isometry computes in closed form:

$$\psi_\beta(\mu_m\,\sigma_t(\mu_m^*)) = m^{-it}\sum_{n:\,m|n}\frac{n^{-\beta}}{\zeta(\beta)} = m^{-\beta-it},$$

entire in t — the multiplicative correlations cross $\operatorname{Im} t = -\frac{1}{2}$ (indeed any strip) with no boundary obstruction, and no ζ-zero appears. □

(iii) Localization (Prop E.3). The ζ-data carrying the zeros enters the BC modular theory only through the **mixed additive×multiplicative correlations** — the $\mathbb{Q}/\mathbb{Z}$ phase generators $e(a/q)$ against the isometries (Ramanujan-sum/$1/\zeta$ values; the Weil pairing itself, as certified in W1, is such a mixed correlation). Hence: *any singular part of the arithmetic state — equivalently any boundary irregularity of the cocycle at* $\operatorname{Im} t = -\frac{1}{2}$ *— is carried exclusively by the additive–multiplicative interface.* □

Reading. The route's two "easy" sectors are now *proven* boundary-regular, and the obstruction is localized to the same interface every flank of the program has converged on — the additive×multiplicative seam (Gauss sums, Jacobi/chirality, the order-3 structure, P∞-B's dichotomy, the cubic). The open step's final coordinates: **the boundary behavior at** $-\frac{i}{2}$ **of the mixed-sector cocycle entries** — Ramanujan-weighted, $1/\zeta$-normalized correlation functions whose strip-analyticity is governed by the zero-free region. The circle is exact: the KMS half-boundary regularity of precisely these entries *is* the critical-line statement, now stated on a single explicit factor with two sectors discharged. Next computation: the $e(a/q)$–μ_m mixed two-point function in closed form and its strip continuation.

7. The mixed two-point function in closed form, and the exact circle (Theorem E.4)

The kernel, computed. The arithmetic (von Mangoldt) state's mixed correlation against a phase generator, smeared along the flow, has strip kernel

$$E^\Lambda_{a/q}(s) \;=\; \sum_{n\ge 1}\Lambda(n)\,e^{2\pi i a n/q}\,n^{-s} \;=\; -\frac{1}{\varphi(q)}\sum_{\chi \pmod q}\chi(a)\,\tau(\bar\chi)\,\frac{L'}{L}(s,\chi) \;+\; (\text{explicit imprimitive terms}),$$

with $s = \beta - it$: **Gauss sums × logarithmic derivatives of Dirichlet L-functions.** The additive–multiplicative interface in closed form — the chirality carriers (τ, P∞-B's order-3 amplitudes) weighting the zero carriers (L'/L), exactly as every flank predicted. For $q = 1$: $E^{\Lambda}(s) = -\zeta'/\zeta(s)$.

Theorem E.4 (the exact circle). The strip continuation of the mixed-sector correlations from real t to the modular half-boundary — $\operatorname{Im} t = -(\beta - \frac{1}{2})$, i.e. $\operatorname{Re} s = \frac{1}{2}$, the KMS half-strip at the critical limit $\beta \to 1^+$ — is regular for all (a, q) **iff** no $L(s, \chi)$ vanishes in $\operatorname{Re} s > \frac{1}{2}$:

$$\boxed{\text{mixed-sector cocycle entries regular at } \operatorname{Im} t = -\tfrac{i}{2} \iff \text{GRH } (q{=}1 : \text{ RH}).}$$

Proof. The kernel's strip singularities are precisely the poles of L'/L, i.e. the zeros ρ with $\frac{1}{2} < \operatorname{Re}\rho \leq \beta$; the Gauss-sum weights are nonvanishing. □

The tier upgrade (E.2 identified). Each hypothetical off-line zero ρ contributes a *specific residue functional* to the arithmetic state — the x^{ρ}-term of the explicit formula, visible to no finite stage of the filtration. **The Takesaki singular part of φ_{Λ} is the off-critical spectrum, term by term:** $\varphi_{\text{singular}} = \sum_{\operatorname{Re}\rho > 1/2}(\text{residue functional at } \rho)$. E.2's "$\varphi_s = 0 \iff$ RH" is thereby upgraded from structural [A] to **identified** — the singular part's atoms are enumerated by the off-line zeros. *[§10.1 erratum: "Takesaki singular part" must be read as the Arveson θ-spectral component; the enumeration of the atoms by off-line zeros stands; the kernel computations of §§6–7 are unaffected.]*

Honest calibration, and what was gained. E.4 is an exact equivalence — necessarily so; anything less would have proven RH tonight. The route's yield is structural and real: *(i)* the sector split, with diagonal and multiplicative sectors **proven** boundary-regular (§6) — the Hypothesis lives only at the interface; *(ii)* the obstruction in closed form — Gauss-sum-weighted L'/L poles; *(iii)* the question relocated into the house where the strongest unexhausted tools live (Araki relative entropy, Uhlmann DPI, Takesaki decomposition, flow of weights, martingale convergence — all type-III-native), with the singular part now an *enumerated* object rather than an abstract one. The Riemann Hypothesis, at the end of this arc: **the arithmetic state of the Bost–Connes factor has no singular part — no residue functional survives at the modular half-boundary — arithmetic hides nothing at its horizon.** The open step is unchanged in strength and utterly changed in address.

8. The two constraints: flow-weight dichotomy and the entropy-growth exponent (E.5, E.6)

E.5 (flow-of-weights constraint; the Arveson dichotomy). Under the scaling flow θ_{λ}, each residue functional is an *eigenfunctional*: $R_{\rho} \circ \theta_{\lambda} = \lambda^{\rho - 1/2} R_{\rho}$ — modulus $\lambda^{\beta_{\rho} - 1/2}$. The arithmetic state decomposes as

$$\varphi_{\Lambda} = \underbrace{R_{\text{pole}}}_{\text{weight } \lambda^{1/2}\text{: the trend, the ample/timelike axis}} + \underbrace{\sum_{\beta_{\rho}=1/2} R_{\rho}}_{\text{unimodular: the normal fluctuations}} + \underbrace{\varphi_{\text{singular}}}_{\text{weights strictly between: in-cone violations}}.$$

So RH is an **Arveson-spectrum dichotomy**: the point spectrum of θ on the arithmetic state is $\{\frac{1}{2}\} \cup i\mathbb{R}$ — *dominant or unimodular, nothing between.* Positivity already forbids weight-*matching* the pole — that is exactly de la Vallée Poussin's edge proof ($\beta = 1$ excluded; the 3-4-1 inequality is the Bochner positivity bound in this frame), giving the operator-algebraic home of the classical zero-free region: **the known $\beta < 1$ is the positivity bound; RH is its rigidity sharpening** — no eigenweights in the open cone interior $(0, \frac{1}{2})$. The Lorentzian reading is exact: intermediate weights are directions strictly inside the light cone off the pole axis; E.5 = P∞'s causality, now as a spectral condition on one state. [A, with the edge case [P]-classical]

E.6 (the entropy-growth exponent; Pinsker direction [P]-shape). Let $c(\lambda) = S(\varphi_{\Lambda} \| \psi; A_{\lambda})$ be the c-function of §1. By Pinsker, $c(\lambda) \geq \frac{1}{2}\|\varphi_{\Lambda} - \psi\|^2_{1, A_{\lambda}}$, and the explicit formula makes the scale-λ L^1-discrepancy

$\gtrsim \lambda^{\beta_\rho - 1/2}$ for any zero with $\beta_\rho > \frac{1}{2}$. Hence **any off-line zero forces polynomial growth** $c(\lambda) \gtrsim \lambda^{2\beta_\rho - 1}$, while under RH the GUE-fluctuation budget gives only polylogarithmic growth ($c \asymp \tau^{O(1)}$ in RG time $\tau = \log\lambda$). The supremum of real parts is read off the c-function:

$$\boxed{\Theta \;=\; \sup_{\rho} \operatorname{Re}\rho \;=\; \tfrac{1}{2}\Big(1 + \limsup_{\lambda} \tfrac{\log c(\lambda)}{\log \lambda}\Big)} \qquad \text{(forward direction Pinsker-proved; converse [A]).}$$

RH = the second law at its critical rate: the entropy of arithmetic relative to free grows polynomially in RG time and no faster. The logarithmic crawl (P∞-D §3) and the entropy clock are the same marginal scaling — the route's three rates (resolution per budget, μ_λ, entropy growth) are one number. *[Historical flag — the "polylogarithmic" RH-side rate stated here was corrected in E.6′ (§10.4): the honest dichotomy is polynomial vs linear-in-RG-time, $c(\lambda) = \kappa \log\lambda$ under RH.]*

Close of the arc. The singular part is enumerated (E.4), its scaling weights are classified (E.5), and its detector is an entropy growth exponent (E.6) — with the classical zero-free region recovered as the positivity edge of E.5, exactly where a true frame should place it. What remains is what always remained — excluding the open cone interior — but it is now posed as a *rigidity* statement for one flow on one state, in the house where rigidity theorems live.

9. The 0–1 form: RH as tail-triviality of the prime tensor product (E.7) *(historical — E.7 as stated is killed in §10, three ways, two by certificate; the surviving statement is E.7′, §10.4)*

(a) The factor is an ITPFI over primes [P, **known**]**.** The BC Gibbs state factorizes, $\psi_\beta = \bigotimes_p \psi_\beta^{(p)}$ ($\zeta = \prod_p (1 - p^{-\beta})^{-1}$); the type III_1 classification *is* the Araki–Woods ratio-set computation: the primes generate a dense subgroup of $\mathbb{R}_+^\times$, so the asymptotic ratio set is everything. The factor's very type records the multiplicative independence of the primes.

(b) Zeros are tail phenomena [P]**.** No finite sub-product of Euler factors vanishes anywhere; a zero of ζ exists only in the infinite assembly — it is a **collective resonance of all primes**, invisible to every finite tensor block. Hence each residue functional R_ρ, and with it the entire singular part $\varphi_{\text{singular}}$ (E.4), is a functional on the **tail** of the prime tensor product.

(c) Conjecture E.7 (the positive-cone modular 0–1 law). *On the prime ITPFI, every θ-finite positive tail functional has scaling weight in $\{\lambda^{1/2}\} \cup \{$unimodular$\}$: tail events carry weight* 1 *(the pole — the trend, probability one) or weight* 0*-type oscillation (the on-line spectrum) — nothing in between.* By E.4–E.6 this is **equivalent to RH**: an off-line zero is exactly a tail event of intermediate weight $\lambda^{\beta - 1/2}$, $0 < \beta - \frac{1}{2} < \frac{1}{2}$.

(d) Why this is the right final form. Kolmogorov's 0–1 law is *the* theorem that converts independence into tail-triviality — and independence of the primes is precisely what the ITPFI structure encodes and what the factor's type already certifies at the ratio-set level. The probabilistic heuristics of the subject (Möbius randomness, Chowla, "the primes flip fair coins," GUE) are all shadows of one exact statement: **the modular-weighted tail σ-algebra of the primes is trivial.** The classical edge ($\beta < 1$, dlVP) is the positivity bound (E.5); the interior exclusion is the 0–1 law the positive cone has never been asked to prove. Strike direction, named: establish a noncommutative 0–1 law for θ-finite positive functionals on ITPFI factors with dense ratio set — the hypothesis (independence + density) is exactly what the primes supply, the conclusion (no intermediate tail weights) is exactly RH, and the toolbox (Araki–Woods, Connes' T/S invariants, tail-triviality of product states, the flow of weights) has never been pointed at this target.

$$\boxed{\text{RH} \;=\; \text{the primes are independent enough that arithmetic has a trivial tail.}}$$

10. The verdict of the assault: E.7 falls, and what it leaves behind (E.8–E.9, E.7′′, E.6′)

Five research flanks plus two adversarial cross-examinations (2026-06-11). The conjecture died honorably — by three independent demolitions, two carrying exact certificates — and the salvage is the sharpest localization the program has produced.

10.1 Three demolitions of E.7 as stated [P, certified]

(i) Well-posedness. A *bounded* positive eigenfunctional of non-unimodular weight is zero: $\varphi(\theta_\lambda(1)) = \varphi(1)$ versus $\lambda^w\varphi(1)$ forces $\varphi = 0$ (two lines; automorphisms are isometric on the dual). So *no* residue functional — pole, on-line, or off-line — is a positive bounded functional; non-unimodular candidates are necessarily unbounded weights, for which "tail-supported" was never defined. E.7 quantified over a class its own objects cannot inhabit.

(ii) Abundance. Positive θ-homogeneous weights of **every** intermediate exponent exist abstractly: Haagerup's L^p cones realize each $c = 1/p \in (0, 1]$ on the core of any III_1 factor; Laplace-tilted ultralimits manufacture tail-supported eigenfunctionals of any prescribed weight with zero arithmetic input; and — the cross-examiner's decisive addition — the **PNT-tilted Banach-limit weight** $\tilde{\varphi}_\sigma$, built *from* Λ *itself*, is positive, tail-supported, θ-homogeneous of modulus $\lambda^{\sigma-1/2}$ for every $\sigma \in (\frac{1}{2}, 1)$, **unconditionally, with no off-line zero** (`probe_e7_cross_exam_tilt_weight.py`, GREEN: $M(2x)/M(x) \to 2^\sigma$, single-block mass fractions $\to 0$). Even "tie the functional to the ledger" fails in its naive form.

(iii) The forward direction. An off-line zero does not produce a positive object at order 1: $R_\rho + R_{\bar{\rho}} \sim \lambda^{\sigma-1/2}\cos(\gamma \log\lambda + c)$ is sign-changing, $\varphi_\Lambda - (1-\varepsilon)R_{\mathrm{pole}}$ is never positive (atomic ledger vs continuous pole), and **Littlewood (1914)** makes the pole-subtracted ledger oscillate at exponent $\frac{1}{2}$ *even under RH.* Linear positivity of Λ provably caps at Landau/de la Vallée Poussin ($\mathrm{Re}\,\rho \le 1$): *E.7 was the classical zero-free-region mechanism extrapolated past its load limit.*

Erratum to §§2, 7 (the Takesaki framing). At $\beta > 1$ the arithmetic state is *normal*; its Takesaki singular part is identically zero. The residue functionals live in the **Arveson θ-spectral decomposition**, not the Takesaki decomposition; §2's E.2 chain and §7's "singular part = enumerated off-line zeros" must be read in the Arveson sense (spectral components of a normal functional under the flow), with the normal/singular dichotomy meaningful only in the critical limit. Recorded as stated-too-strongly; the closed-form kernel computations of §§6–7 (E.4) are unaffected.

10.2 Theorem E.8 (the fake curve; strategy-closure) [P, exact-arithmetic certificate]

There is a Beurling-type arithmetic system over the value group $q^{\mathbb{Z}}$ ($q = 9$) with zeta function

$$Z(T) = \frac{1 - 7T + 9T^2}{(1-T)(1-9T)},$$

having: integer point counts with $M_d \ge 0$ for all d (verified to $d = 300$), Dold integrality, a self-dual functional equation ($\alpha_1\alpha_2 = q$), a positive prime ledger, a rank-2 tail carrier — and generating the **same hyperfinite $\mathrm{III}_{1/q}$ ITPFI with the same discrete scaling flow** as a genuine curve — **whose zeros are off-line** (weights 5.303 and 1.697 against $\sqrt{q} = 3$). Since $|a_1| = 7 > 2\sqrt{q}$, Honda–Tate denies any actual curve: equivalently, *no positive polarization compatible with the q-adjoint relation exists on this carrier.* (`probe_E7_ff_beurling_counterexample.py`, GREEN; independently re-verified by the cross-examiner to $n = 120$.)

Consequence (the No-Go). Independence, ratio-set density, factor type, the ITPFI 0–1 law, the functional equation, integrality, and ledger-positivity — *the entire abstract hypothesis set* — coexist with off-line zeros. **No proof of RH can proceed from these ingredients alone.** This closes the operator-algebraic/probabilistic abstract route by certificate, joining the program's No-Go family (no finite witness; no finite modular witness; no order-2 finite witness) with: *no von-Neumann-abstract witness.* (Scope note

from cross-examination: the fake system is a different ITPFI, and tail structure is not a vN invariant — so this is strategy-closure, not falsity of the $\mathbb{Q}$-statement itself.) This is the Beurling-prime phenomenon (Diamond–Montgomery–Vorhauer) in tail clothing.

10.3 Theorem E.9 (the function-field mechanism, disentangled — and the three obligations over $\mathbb{Q}$)

Over function fields the exclusion of intermediate weights is **the polarization positivity** — Rosati positivity from the theta divisor, equivalently Hodge index/Castelnuovo–Severi on $C \times C$, paid for by Riemann–Roch on the surface — and *not* finite-dimensionality, not the 0–1 law, not factor type, not the FE pairing, not Λ-positivity (each excluded by E.8). But the cross-examination caught a decisive subtlety: the "3-line rigidity" (positive form $+FF^\dagger = q \cdot 1 \Rightarrow |\alpha| = \sqrt{q}$) is **false in infinite dimensions** — $F = \sqrt{q}\, S^*$ (shift adjoint) satisfies $FF^\dagger = q \cdot 1$ with eigenvalues filling the open disc. The correct template over $\mathbb{Q}$ has **three obligations**, of which finite rank silently discharged two: 1. **a positive-definite FE-compatible form on the (infinite-dimensional) tail carrier** — over $\mathbb{Q}$ this *is* Weil positivity, the known wall; 2. **the strong adjoint relation** $F^\dagger F = q \cdot 1$ (the isometry direction — then $\|Fv\|^2 = q\|v\|^2$ pins the point spectrum in one line), not merely $FF^\dagger = q \cdot 1$; 3. **surjectivity / Fredholm index 0 of** $F/\sqrt{q}$ to control the approximate spectrum — free in finite rank, an honest *new named obligation* over $\mathbb{Q}$.

10.4 E.7″ (the surviving statement) and the corrected entropy dichotomy (E.6′)

E.7″. The repair moves positivity one order up, where it belongs: with $Q(f) = (\varphi_\Lambda - R_{\text{pole}} - R_{\overline{\text{pole}}})[|\int f(\lambda)\theta_\lambda(a) d^\times\lambda|^2]$ the scaling two-point form of the pole-subtracted arithmetic state, each on-line zero contributes a PSD rank-1 block and each off-line FE-pair an *indefinite rank-2 block*; the conjecture is $Q \geq 0$ on the test cone — **which is the Weil positivity criterion transported to the BC scaling frame**, classical, RH-equivalent. The assault thus lands exactly on P∞-C's pillar IV: the polarization is the effectivity/Riemann–Roch step of the Castelnuovo map. The order-1→order-2 jump is the program's oldest theorem (CBS: order 2 carries what order 1 cannot) arriving from a fifth independent direction — now with *four* independent proofs that nothing linear or abstract suffices (Littlewood; Landau cap; Haagerup abundance; the fake curve).

E.6′ (corrected exponents). The honest dichotomy is **polynomial vs linear-in-RG-time** (not polylog): under RH, $c(\lambda) = \kappa \log\lambda\,(1 + o(1))$ — Cramér's theorem gives an $O(1)$ entropy budget per octave with $\sigma_0^2 = \sum_\rho 1/(\rho(1-\rho)) = 2 + \gamma_E - \log 4\pi \approx 0.0461$, and the zero tower renormalizes only the slope; an off-line zero at β forces $c(\lambda) \geq \lambda^{2\beta-1-o(1)}$ along a subsequence. With $L = \limsup \log c / \log\lambda$: $\Theta = \frac{1}{2}(1 + L)$, the $\geq$ direction **unconditional** — so $c(\lambda) = \lambda^{o(1)} \Rightarrow$ RH stands as an unconditional criterion, with the pole-matching normalization (which of three normalizations defines c) flagged as the load-bearing bookkeeping.

10.5 What the program keeps

(1) The **free lemma**: normal-level tail triviality for the non-tracial BC product state (three literature routes; apparently unstated for BC) — every residue functional is *necessarily* outside the normal predual's reach. (2) The **novelty confirmation**: the explicit-formula-as-eigenfunctional-decomposition of φ_Λ on the BC factor under the scaling flow appears genuinely new as a framing (closest precursor: Connes 1999, off-line zeros as resonances). (3) The **No-Go by certificate** (E.8). (4) The **three-obligation template** (E.9) — obligations 2 and 3 are new, named, and invisible in every finite-rank intuition. (5) The **unconditional entropy criterion** (E.6′). (6) Total consilience: the assault, launched from the opposite side of the program (probabilistic/entropic), landed on the same single residual — the polarization positivity of P∞-C pillar IV — through its own independent chain. The wall is one wall; we have now surveyed every face of it.

11. The strike on obligations (2) and (3): the Krein category collapses them off the wall

The assault's net asset (E.9) was to split the function-field exclusion into **three** obligations for the $\mathbb{Q}$ tail carrier — (1) the positive FE-compatible form (Weil positivity, the wall); (2) the **strong adjointness** $F^\dagger F = q \cdot \mathbf{1}$; (3) **index-0/surjectivity** of $F/\sqrt{q}$ — and to observe that obligations (2),(3) were *silently free* in finite rank but are not obviously free in ∞ dimensions (the cross-examiner's $F = \sqrt{q}\, S^*$ fills the open disc, killing naive rigidity). The strike question is whether (2),(3) are nonetheless provable for $\mathbb{Q}$ **independently of (1)**. They are — in the Krein/indefinite-metric category that is the correct ambient for the explicit formula. The point is that (2),(3) are *structural* (they hold whether or not RH does), so only (1) carries RH; the apparent three-obligation barrier was two-thirds illusory.

Fix the Weil pairing on test functions f (supported in $|u| \le L$), written $F(s) = \int f(u)\, e^{(s-1/2)u}\, du$ and

$$B(f,g) \;=\; \sum_{\xi\text{-zeros }\rho} F(\rho)\,\overline{G(1-\bar\rho)} \;=\; G(\varphi_{f,g}), \qquad \varphi_{f,g}(u) = \int f(u+v)\,\overline{g(v)}\,dv,$$

with $G(\varphi)$ the **geometric (Weil) functional** — pole terms $\Phi(0)+\Phi(1)$, prime sum $-\sum_n \Lambda(n) n^{-1/2}[\varphi(\log n) + \varphi(-\log n)]$, archimedean term $-\varphi(0)\log\pi + \frac{1}{2\pi}\int h(t)\,\mathrm{Re}\,\psi(\frac{1}{4}+\frac{it}{2})\,dt$. The carrier is $\mathcal{H} = \overline{\{f\}}$ with indefinite inner product B; $W = d/du$ is the scaling generator; $\iota f(u) = f(-u)$ is the FE involution; J is complex conjugation; $C = J\iota$.

Theorem E.10 (obligation (2) is unconditional: W is B-skew). $B(Wf,g) + B(f,Wg) = 0$ for all f, g — equivalently $W^{\dagger_B} = -W$ in the Krein metric — and the identity holds **on the geometric side, with no use of the zeros or of RH.**

Proof. $B(Wf,g)+B(f,Wg) = G(\varphi_{Wf,g}+\varphi_{f,Wg})$, and $\varphi_{Wf,g}+\varphi_{f,Wg} = \partial_u \int f(u+v)\overline{g(v)}dv - \int f(u+v)\overline{g'(v)}dv$; integrating the second by parts, $\int f(u+v)\overline{g'(v)}\,dv = \partial_u \varphi_{f,g}$, so the cross-correlation **vanishes identically as a function of** u, hence $G(\cdot) = 0$ term-by-term (pole, prime, arch each annihilate the zero function). □

Thus the "strong adjointness" $F^\dagger F = q \cdot \mathbf{1}$ — the exact-isometry obligation — is, in the additive scaling presentation, the statement that the generator is B-skew (its Cayley transform a B-isometry). It is a **clean identity of the explicit formula**, true unconditionally. Obligation (2) never touched RH.

Theorem E.11 (obligation (3) is unconditional: the index vanishes by FE conjugation). The FE involution is a B-isometry, $B(\iota f, \iota g) = \overline{B(g,f)}$; consequently $C = J\iota$ is an antilinear B-isometry with $C^2 = \mathbf{1}$ and $CW = -WC$. By von Neumann's theorem an operator commuting with a conjugation that anti-intertwines it has **equal deficiency indices**, so W admits B-skew-self-adjoint extensions and the Fredholm index of $W \pm \frac{1}{2}$ (equivalently of $F/\sqrt{q}$ in the multiplicative picture) is 0. The index obstruction is therefore **automatically discharged by the functional equation**, with no input from positivity.

Proof of the isometry. $\varphi_{\iota f,\iota g}(u) = \int f(-u-v)\overline{g(-v)}\,dv = \overline{\varphi_{g,f}}(u)$ after $v \to -v$ and using realness of the geometric kernel; G commutes with this conjugation because the pole pair $\{0,1\}$, the prime kernel ($\log n \leftrightarrow -\log n$ symmetric), and $\mathrm{Re}\,\psi$ are all reflection-even. □

Theorem E.12 (the residue: $\kappa = 1$, and RH as a Pontryagin index). With (2),(3) discharged structurally, the *entire* RH content sits in obligation (1): the zeros-only form B is positive semidefinite iff RH. Equivalently, the **pole-extended** form $B_{\mathrm{full}} = B + \Phi(0)\overline{\Phi(1)} + \Phi(1)\overline{\Phi(0)}$ has **exactly one negative square** — $\kappa(B_{\mathrm{full}}) = 1$, the hyperbolic cell of the pole at $s = 1$ — and RH $\iff$ the carrier $(\mathcal{H}, B_{\mathrm{full}})$ is a Pontryagin space Π_1 (one negative square, no more). On-line zeros give PSD rank-1 blocks; an off-line FE-quadruple would add a second negative square, $\kappa \ge 2$. This is Weil positivity restated as **index rigidity at** $\kappa = 1$.

Certificate. `empirical/p_infinity/probe_obligations_2_3_krein.py` (GREEN, deterministic, local): CHECK 0 validates the geometric functional against the first 120 ζ-zeros (rel err 2.9×10^{-4}); CHECK 1 certifies Theorem E.10 — $B(Wf,g)+B(f,Wg) = 0$ to $\sim 10^{-13}$ relative, with the cross-correlation cancelling

pointwise to $\sim 10^{-15}$ **before any zero is summed**; CHECK 2 certifies the E.11 isometry $B(\iota f, \iota g) = \overline{B(g,f)}$ to $\sim 10^{-12}$; CHECK 3/4 build the 8×8 Weil Gram, confirm Hermiticity (3×10^{-12}), the zeros-only form PSD to the certified quadrature floor (Weil-positivity witness — a genuine negative square here would *disprove* RH), and $\kappa(B_{\text{full}}) = 1$ exactly.

What this buys. Two of the three obligations the assault manufactured are now **theorems, unconditional, independent of (1)** — the strong-adjointness and index obstructions are artifacts of the indefinite metric and the FE, not real barriers. The barrier is, with full rigor, *exactly one scalar inequality*: $\kappa = 1$ rather than $\kappa \geq 2$. The Krein category is the right home for the explicit formula precisely because it makes (2),(3) free and isolates RH as **index rigidity of a Pontryagin space** — the sharpest statement of the single residual the program has produced, and the cleanest dictionary entry yet for P∞-C's pillar IV. *(Open, unchanged in strength: no off-line quadruple, i.e. $\kappa \leq 1$.)*

12. The N_κ correspondence: the index made classical, and what the lens does and does not give

Section 11 isolated RH as Pontryagin index rigidity ($\kappa = 1$ for the pole-extended form; $\kappa = 0$ after removing the pole). That abstract index has a *classical analytic realization*, and the generalized-Nevanlinna (N_κ) apparatus of Krein–Langer is the structure theory built precisely for it. This section records the correspondence, its certificate, the honest gap, and the genuinely new ground it opens.

Setup. Let $\Xi(z) = \xi(\frac{1}{2} - iz)$, real-entire, even, order 1, pole-free; a zero $\rho = \beta + i\gamma$ of ζ maps to $z_\rho = -\gamma + i(\beta - \frac{1}{2})$, so *on-line* $\iff$ *real* z_ρ, and an off-line quadruple $\{\beta \pm i\gamma,\ 1-\beta \pm i\gamma\}$ places **exactly two** points in $\mathbb{C}^+$. Set $\widetilde{N}(z) = -\Xi'/\Xi(z) = i\,(\xi'/\xi)(\frac{1}{2} - iz)$.

Proposition E.13 (RH as Nevanlinna class membership). Since $\operatorname{Im}[\,1/(z-t)\,] = -\operatorname{Im} z/|z-t|^2$ has fixed sign on $\mathbb{C}^+$ for real t, the Hadamard expansion of Ξ'/Ξ over paired zeros $\pm z_k$ gives: **RH** $\iff$ **$\widetilde{N}$ is Herglotz (class N_0)** $\iff$ **the Nevanlinna kernel** $K(z,w) = (\widetilde{N}(z) - \overline{\widetilde{N}(w)})/(z - \bar{w})$ **is PSD on $\mathbb{C}^+$.** More precisely (Krein–Langer), $\widetilde{N} \in N_\kappa$ with $\kappa = \#\{\text{zeros of } \Xi \text{ in } \mathbb{C}^+\}$: each off-line quadruple costs exactly 2 negative squares (a hypothetical zero on the real axis of s with $\gamma = 0$ would cost 1). This is §11's E.12 realized on a classical object — the $\kappa = 1$ pole cell of B_{full} was ζ's pole at $s = 1$; on the pole-free Ξ the statement is the cleanest possible: RH $\iff$ $\kappa = 0$.

What Krein–Langer delivers. (a) **Factorization:** $N \in N_\kappa$ iff $N = (r^\#/r)\,N_0$ with $N_0 \in N_0$ and r a degree-κ polynomial vanishing at the generalized poles of non-positive type — a *finite half-plane Blaschke factor*. The Weil-positivity defect polynomial is hereby **named**: it is the Blaschke numerator, and RH says the inner factor is trivial. (b) **Operator realization:** N_κ functions are exactly the Weyl/Q-functions of self-adjoint relations in Π_κ, and a Π_κ self-adjoint operator has **at most κ non-real eigenvalues** — positivity-of-a-form traded for a *bound on an integer*. (c) **Representation:** the Herglotz measure becomes positive after multiplication by $|r(t)|^2$ — the defect parametrized at degree 2κ.

The honest gap (recorded with both hands). N_κ is a *finite-κ theory*. That Ξ'/Ξ lies in N_κ for some finite κ — i.e. that off-line zeros are finite in number — is itself **open**, weaker than RH and stronger than what is known. And Krein–Langer is *sign-preserving, not sign-creating* (Toland–Singer class, cf. N15 §3.1): counting negative squares does not vanish them. This section is a reformulation with new tooling, not a closure; the residual ($\kappa = 0$, = pillar IV) is unchanged in strength.

External corroboration (Suzuki 2026, screw function). The $\kappa = 0 \iff$ RH statement of this section is independently published: Suzuki's screw function g associated with ζ (arXiv:2606.09096, building on Su23) satisfies "g is a Krein–Langer screw function $\iff$ RH" — the *same* N_κ/Krein–Langer criterion, arrived at independently. Moreover his Theorem 5 constructs the finite-stage Hilbert–Pólya operator (deficiency indices $(1,1)$, unconditional real zeros) that is the concrete de Branges realization of §11's E.10–E.12 (our abstract Krein proof). The full weld — his screw function = our Weil form, certified to 0.78%, and the

consequent discharge of (H1) from the program's ledger — is developed in **P∞-F §12**.

The new ground (three items, none available in the positivity frame). (i) κ **is a winding number**: by the argument principle the index equals a contour integral of Ξ'/Ξ over ∂(half-plane window) — which is the Backlund–Turing zero-counting object, i.e. **the Pontryagin index is detected by** $S(T)$, with its own analytic toolkit (Selberg CLT, moments of $S(T)$, Montgomery) — a *different hard object* than a positive-definiteness proof. (ii) **Index transport replaces positivity transport** as the $\mathbb{F}_q \to \mathbb{Q}$ bridge: over function fields $\kappa = 0$ is a theorem (Weil/Hodge), and the question becomes whether the *integer* index can be pinned along a deformation from the $\mathbb{F}_q$ anchor — homotopy/spectral-flow/Fredholm-determinant tools, a category never aimed at this residual by the program. (iii) **The rigorous home exists**: Pontryagin de Branges spaces $\mathcal{H}(E)$ with finite negative index (Kaltenbäck–Woracek) dock directly onto the program's de Branges thread (N0 §2 ≅ Connes Thm 6.1), with Ξ as Hermite–Biehler structure function and κ its negative index.

Certificate. `empirical/p_infinity/probe_nkappa_krein_langer.py` (GREEN, deterministic, local, mpmath dps = 25–30): CHECK A — the Nevanlinna kernel of $-\Xi'/\Xi$ on 21 points of $\mathbb{C}^+$ for the **true** ζ is PSD (min eigenvalue $+9.4 \times 10^{-26}$ at scale 2.2; $\kappa = 0$, a genuine negative square would *disprove* RH). CHECK B — counting law on rational models built from the first 12 true zeros: all-real $\to \kappa = 0$; imaginary pair $\pm i\delta \to \kappa = 1$ exactly; off-line quadruple $\to \kappa = 2$ exactly. CHECK C — a fake off-line quadruple $w = 10 + 0.25i$ **grafted onto the true** ζ produces exactly 2 negative squares. CHECK D — the winding $(1/2\pi i) \oint \Xi'/\Xi \, dz$ over $[-20, 20] \times [0.1, 2]$ returns 0.000000 (true) and 2.000000 (grafted): **index = argument-principle count = the** $S(T)$ **object**, machine-verified.

13. The $S(T)$ face: the index is the Turing discrepancy, and partial rigidity is already a theorem

§12(i) promised that the Pontryagin index is detected by $S(T)$. Made precise, the detection splits into three exact identities — and the import runs *both* ways: the index lens reorganizes a century of zero-density and verification literature as a **graded partial-rigidity theory** with one open corner.

Proposition E.14 (the $S(T)$ **dictionary).** (a) *The index is the Turing discrepancy*: for a window W of ordinates, $\kappa(W) = N(W) - N_0(W)$, where $N(W)$ is the all-zero count (argument principle / $S(T)$, blind to β) and $N_0(W)$ the Hardy-style sign-change count of Ξ on the real z-axis (which sees only on-line zeros of odd order). The discrepancy equals κ plus twice the number of even-order on-line zeros — both are "non-simple events", and the even-order case is exactly the $d \to 0$ marginal limit below. **This is the quantity zero-verification has computed to be** 0 **to height** 3×10^{12} (Platt–Trudgian): the largest exact index computation in existence, reread. (b) *Resolution grading*: the winding of Ξ'/Ξ over $[T_1, T_2] \times [\varepsilon, 1]$ in $\mathbb{C}^+$ counts exactly the zeros with displacement $\beta - \frac{1}{2} > \varepsilon$ — $\kappa_\varepsilon(W)$, computable without touching the critical line. (c) *Resolution-cost law (measured)*: the most negative Nevanlinna-kernel eigenvalue for a displacement-d quadruple seen from test height h scales as $|\lambda_-| \sim d^{2.2}\, h^{-5.3}$ — quadratic marginality in displacement (a quadruple splitting off the line is a second-order event), steep power cost in distance.

Imported partial rigidity (literature, reorganized; recalled not certified). $\kappa_\delta(T) := N(\frac{1}{2} + \delta, T)$ obeys: Bohr–Landau (1914): $O_\delta(T)$, hence **density zero at every fixed resolution**; Selberg (1946), *uniform in* δ: $\kappa_\delta(T) \ll T^{1-\delta/4} \log T$; Guth–Maynard (2024): $\ll T^{30(1-\sigma)/13+o(1)}$, the current record at large displacement; and the critical-line proportion ($\geq 2/5$ Conrey; $\approx 41.7\%$ current) gives the global $\kappa(T) \leq 0.583\, N(T)$ unconditionally. In the index language: **the index is already known to be density-zero rigid at every positive resolution, with a uniform graded rate; RH is exact rigidity at the** $\delta \to 0$ **corner.** Note the consilience with P∞-D: inverting Selberg's uniform exponent, resolving displacement δ at height T costs budget $\sim T^{\delta/4}$, i.e. $\delta \sim 4 \log B / \log T$ — *the same logarithmic resolution-per-budget law as the crawl* (P∞-D §3; Śliwiński form). The $S(T)$ face and the apex face crawl at the same rate, and it is the program's marginality.

Honesty clause. Selberg's CLT for $S(T)$ controls the *fluctuations of* N and is structurally blind to the

$N - N_0$ split — moment control of S does not bound κ, and we record that the CLT route, as such, dies here (the fluctuation budget $\sqrt{\log\log T}$ is a context, not a constraint). The lens manufactures no new bound; its yield is the *organization* (graded index rigidity, with the open corner localized at uniform-$\delta \to 0$) and the dictionary (a), which converts every future verification run into an exact index computation. A second structural remark, flagged [A]: off-line zeros force **double ordinates** (ρ and $1-\bar{\rho}$ share γ), so the index is also bounded by the failure of level repulsion — the GUE face (cf. the gue_cumulant arc) of index rigidity; under GUE statistics double ordinates are second-order suppressed, matching the measured $d^{2.2}$.

Certificate. `empirical/p_infinity/probe_index_resolution_sT.py` (GREEN): ε-graded winding counts $\{0, 2\}$ exactly as the floor passes the grafted displacement ($\varepsilon = 0.05, 0.15$ see the quadruple, 0.35 does not; true ζ: 0 at all floors); Turing discrepancy: true $N = 1.0000$, $N_0 = 1$, $\kappa = 0$; grafted $N = 3.0000$, $N_0 = 1$, $\kappa = 2$ — the index read off as $N - N_0$ on the nose; resolution-cost law measured over $d \in [0.05, 0.4]$, $h \in [0.5, 4]$ with clean monotonicity in both arguments.

14. The spectral-flow formulation: index transport along the q-deformation, and Krein's collision law

§12(ii) proposed replacing positivity transport by **index transport** across the $\mathbb{F}_q \to \mathbb{Q}$ bridge. The correct machinery is Krein–Pontryagin stability theory, and it delivers a sharp mechanism statement plus an exact identification on the function-field side.

Theorem E.15 (Krein-collision rigidity). (a) *(Pinning; classical Pontryagin, certified.)* A B-self-adjoint operator in Π_κ has at most κ conjugate pairs of non-real eigenvalues. In Π_1 with the pole pair occupying the single slot, **every other eigenvalue is pinned to the real axis under arbitrary continuous deformations** — real-eigenvalue collisions pass through; they cannot leave. (b) *(Jump locus.)* Along a continuous family of forms and B_τ-self-adjoint operators with $\kappa(B_\tau)$ finite, the index can change **only through a degeneracy of the form** (the definite cone collapsing through its boundary); away from degeneracies κ is locally constant. [P] in finite rank; in infinite dimensions under gap-continuity hypotheses [A]. (c) *(The $\mathbb{F}_q$ anchor: Honda–Tate = Krein collision; certified exactly.)* On the rank-2 carrier $F = \text{companion}(\lambda^2 - a\lambda + q)$, the invariant-form equation $F^{\mathsf{H}} B F = qB$ admits a positive-definite solution iff $|a| < 2\sqrt{q}$; at $|a| = 2\sqrt{q}$ the definite cone degenerates (Jordan collision on $|\alpha| = \sqrt{q}$); for $|a| > 2\sqrt{q}$ *every* invariant Hermitian form is hyperbolic. The eigenvalues leave the Weil circle exactly there: **the Honda–Tate boundary is the Krein collision locus**, and the witness $w(a)$ (max-min-eigenvalue over the invariant cone) crosses zero at $a = 2\sqrt{q}$ linearly — the marginality measured. E.8's fake curve ($a = 7$, $q = 9$, $w = -0.031$) sits past the collision: its non-realizability as a curve *is* Krein instability. (d) *(Application to ζ;* [C]*.)* The CCM λ-family (2511.22755: rank-one-perturbed scaling triple, self-adjoint at every λ) is a path of $\kappa = 0$ objects whose boundary limit is Ξ; RH = **no index jump at the boundary**. By (b) the jump mechanism is classified: it requires either a form degeneracy along the path — excluded at finite λ by the program's nondegeneracy results (adelic Legendre/Tate self-duality, probe GREEN) — or loss of resolvent control *in the limit*. The open step is thereby re-tooled, not re-weighed: **"prove the limit inequality" becomes "prove the convergence $k_\lambda \to \Xi$ in a topology under which the integer κ is continuous"** — the norm-resolvent vs strong-resolvent dichotomy, a fully classified phenomenon in perturbation theory, aimed for the first time at the program's single residual. Strong-resolvent convergence permits index jumps; norm-resolvent-type control forbids them. CCM's missing Lemma 7.2 *is* the statement that the convergence is of the controlled kind.

Certificate. `empirical/p_infinity/probe_krein_collision_spectral_flow.py` (GREEN, seeded, exact): Π_1 never exceeds one non-real pair over 20,000 samples (Π_2: two, achieved 89.5%); 40 random Π_1 deformation paths $\times$ 2001 steps with the pole pair present: zero excursions — pinning holds; the Honda–Tate sweep $a \in [0, 8]$, $q = 9$: $w(a) > 0$ below 6, $w(6) = 0.0000$, $w(a) < 0$ above, single sign change located at $a = 6.000$; $|\alpha|$ splits off $\sqrt{q} = 3$ exactly at the collision.

15. The rank-one determinant: the open step as one weighted uniform bound (Theorem E.16)

§14(d) re-tooled the residual as a convergence-topology question for the CCM λ-family. The family's defining feature — the Dirac-type operator is a **rank-one** perturbation of the self-adjoint scaling operator — makes the relevant determinant *explicit*, and the Krein/Aronszajn–Donoghue formula turns the whole architecture into three short statements and one quantitative spec.

Theorem E.16. Let A be self-adjoint, $B_t = A + t\langle\varphi, \cdot\rangle\varphi$ the rank-one flow, $F(z) = \langle\varphi, (A-z)^{-1}\varphi\rangle$ its Borel transform, and $D_t(z) = 1 + tF(z)$ the perturbation determinant ($= \det(B_t - z)/\det(A - z)$ in finite rank; certified to 10^{-9}). **(a) (Herglotz reality — every finite stage is pinned.)** F is Herglotz, so $D_t(z_0) = 0$ with $\operatorname{Im} z_0 > 0$ would force $F(z_0) = -1/t \in \mathbb{R}$, impossible for non-constant Herglotz F (open mapping): **all zeros of D_t are real for every t**, with Aronszajn–Donoghue interlacing. Renormalization $c_\lambda^{-1} D_\lambda$ changes no zeros. The reality of the spectrum along the *whole* deformation is structural, free, and certified. **(b) (Hurwitz transport — the closing mode.)** If the renormalized determinants converge **locally uniformly on a complex domain** Ω to a non-degenerate limit, the limit has no zeros in $\Omega \setminus \mathbb{R}$. Hence RH-in-a-box follows from locally uniform convergence on the box. (Positive-mode certificate: the genus-0 Hadamard truncations $p_N(z) = \prod_{k \le N}(1 - z^2/\gamma_k^2)$ converge to $\Xi(z)/\Xi(0)$ on complex boxes at the predicted $|z|^2 \cdot \mathrm{tail}(N)$ rate.) **(c) (Montel reduction.)** By (a)+(b) and Montel–Vitali, the *entire* open step reduces to: **one locally uniform upper bound on the renormalized family $\{c_\lambda^{-1} D_\lambda\}$, plus non-degeneracy of the limit, plus identification of the limit with Ξ on $\mathbb{R}$.** A normal family of real-rooted functions converging on $\mathbb{R}$ to Ξ converges locally uniformly to Ξ and RH follows. The residual is *tightness* — no escape of zero-mass — for an explicitly given Herglotz-built family. **(d) (The weighted-tail obligation — where weak convergence dies; certified.)** Real-axis-uniform closeness does **not** substitute for (c): the shift $H = \Xi + \delta$ with $\delta = -1.05\, E_{5,6}$ ($E_{5,6}$ = the extremum of Ξ between γ_5, γ_6), of size $|\delta|/\sup|\Xi| = 4.9 \times 10^{-9}$ *uniformly on all of* $\mathbb{R}$, migrates two zeros off the axis: argument-principle count on $[34, 36.5] \times [-1, 1]$ goes $0 \to 2$, the pair located at $34.17 \pm 0.33i$. The protection threshold at height T is the **local envelope** $|\Xi| \sim \mathrm{poly}(T)\, e^{-\pi T/4}$ — the archimedean Γ-decay. The convergence obligation is therefore *weighted*:

$$\sup_{|t| \le T_\lambda} \; e^{+\pi|t|/4} \left|\hat{k}_\lambda(t) - \Xi(t)\right| \;\longrightarrow\; 0, \qquad T_\lambda \to \infty,$$

and *no unweighted real-axis norm, however small, suffices.* This sharpens the missing CCM Lemma 7.2 into a falsifiable spec: **the convergence must beat the archimedean redshift in the tail** — the same $e^{-x/2}$-family envelope that appears as P∞-D's crawl weight and N3/N4's reflection kernel. The exp-small marginality of the program's earlier probes is now identified as the *exact price of zero-protection.*

Status. (a), (b), (c) are [P] at the architecture level (and certified); the application to the specific CCM objects is [C] pending their exact normalizations. What is new: the localization of the entire residual to a single weighted uniform bound on an explicit family, the proof that the reality mechanism costs nothing at every finite stage, and the spec (d) that tells any future convergence proof *exactly which norm it must control and why weaker ones provably fail. Scope of (c)–(d) (2026-10-10).* For families already known to be real-rooted, such as the CvS/CCM ground states under the simple-even hypothesis, the locally uniform upper bound of (c) costs nothing. The Hadamard–Laguerre bound $|F(z)| \le e^{S|z|^2}$, $S = -F''(0)/2$, holds with S controlled by convergence at a single real point below the first zero. So convergence on one real segment already gives the full conclusion, and the weighted obligation (d) binds only families not known to be real-rooted, such as $\hat{k}_\lambda$ itself (N7 Thm. 5.2 and Cor. 5.3–5.5, with the Davenport–Heilbronn control in N7 Obs. 5.7–5.8).

Certificate. `empirical/p_infinity/probe_rank_one_determinant_spec.py` (GREEN): Krein identity det-ratio $= 1 + tF$ at 60 random (t, z) to 10^{-9}, $\operatorname{Im} F > 0$ on $\mathbb{C}^+$, perturbed spectra real and interlaced; Hadamard truncations $N = 60 \to 120$ converging locally uniformly on $|z| \le 5$ with monotone error; the 4.9×10^{-9}-uniform shift migrating exactly two zeros to $34.17 \pm 0.33i$ (winding $0.0000 \to 2.0000$, residual $|H| = 3 \times 10^{-39}$ at the located pair).

Anchors (§§1–15). P∞-C §3 (the factor and trace-scaling); P∞-D §§3,7 (crawl, entropic tactic; log resolution-per-budget law; the $e^{-x/2}$ envelope); N14 (Weil = relative entropy); N14 §7.4 (channels, entropy production); CCM 2511.22755 (stage positivity; Lemma 7.2 convergence; rank-one structure); Connes 2602.04022 §6.5 (prolate convergence, Hurwitz architecture); Araki (relative entropy in von Neumann algebras); Uhlmann (DPI); Takesaki (decomposition; conditional expectations); Ohya–Petz, Hiai (martingale convergence of relative entropy); Krein–Langer 1977 (N_κ, Π_κ operator theory; factorization); Krein (stability of eigenvalues on the circle; signature collisions); Aronszajn–Donoghue (rank-one spectral theory); Kaltenbäck–Woracek (Pontryagin de Branges spaces); von Neumann (deficiency indices and conjugations); Backlund/Turing ($S(T)$ zero counting); Bohr–Landau 1914, Selberg 1946, Guth–Maynard 2024 (zero-density = graded index bounds); Platt–Trudgian (verification to 3×10^{12} = exact index computation); Conrey, PRZZ (critical-line proportion); Honda–Tate (realizability); Montel–Vitali, Hurwitz (normal families, zero transport); Paper 32 §4.3 (closure defect = beta source).

16. Interlude: the semantic dictionary of §§11–15

Sections 11–15 were driven by certificates; this section records what they *mean*. The program's working method (Paper 35) holds that the analytical sense-making of the empirical workflow is the actual harvest: each discipline whose stability theory we now match contributes both a confirmation that the structure is real and an intuition for the next move. We flag each entry [**exact**] — a mathematical identification, theorem-level — or [**structural**] — a shared mechanism, transferable but not an isomorphism. The honesty rule of §12 stands throughout: none of these dictionaries closes the residual; their value is that each names a *toolkit* that has never been aimed at it.

16.1 One sentence, six languages

The day's net theorem — *every finite stage of the deformation is rigid for free; the entire content of RH is an integer that must not jump at one boundary point, and protection at that point costs exactly the archimedean envelope* — reads natively in six fields:

Discipline	The object	RH becomes	What imports
Hamiltonian mechanics	Krein signature of eigenvalues	no parametric resonance: opposite-signature collision never fires	Krein–Gelfand–Lidskii stability theory
Quantum field theory	indefinite-metric (Gupta–Bleuler) state space	unitarity of the physical sector: exactly one ghost, decoupled	BRST/ghost-decoupling arguments
Condensed matter	integer topological invariant	topological protection: no phase transition along the q-deformation	gap-closing / spectral-flow analysis
Control engineering	positive-real (passive) impedance	the arithmetic network is passive: no hidden active element	Brune–Foster synthesis; Nyquist criterion
Scattering theory	resonances of a one-port coupling	all states bound, none decaying: no complex resonance	Weyl m-function, resonance-width estimates
General relativity	redshifted signal through a horizon	the convergence proof must communicate at the redshifted amplitude	weighted/conformal estimates at the boundary

The point of the table is not decoration. Six stability theories, each with a century of technique, are

now certified to share their *mechanism* with the residual — and mechanisms, unlike metaphors, transport theorems.

16.2 Time, ghosts, and the one negative square [exact at the structural core]

E.12's $\kappa = 1$ closes a loop the program opened in P∞ itself. The Lorentzian intersection metric of P∞ — signature $(1, \rho-1)$, archimedean place timelike, RH as light-cone causality — *is* the Pontryagin Π_1 structure of B_{full}: **one negative square = one timelike direction = the pole = the archimedean place.** The identification is now operator-theoretic, not metaphorical: the pole pair $\{0, 1\}$ is the unique forced hyperbolic cell, and RH says the geometry is *exactly* Minkowski — no second time direction anywhere in the spectrum. The physics twin of RH in this dictionary is the question "why is there exactly one time dimension?", and the Krein answer is the same in both: a second negative square is an instability, and stable worlds do not have one.

The QFT reading sharpens this. An indefinite-metric state space with one unphysical direction that decouples from the physical sector is not pathology — it is *gauge theory* (Gupta–Bleuler electrodynamics quantizes the photon exactly this way). The pole is the arithmetic ghost: present in the covariant description, absent from physical amplitudes (Weil positivity is stated on the pole-orthogonal subspace). RH = the physical S-matrix is unitary. This dovetails with the metaplectic thread (Q8vM3: Uniform Place Treatment forces Bargmann–Fock at $v = \infty$): the archimedean place was already the program's gauge slot.

16.3 Stability and the Krein budget [exact]

Krein collision — non-real eigenvalues born only from collisions of opposite-signature real/circle eigenvalues — is *the* onset-of-instability mechanism of Hamiltonian mechanics: parametric resonance, the Benjamin–Feir instability of water waves, half-integer resonances in particle accelerators. E.15(c) certifies that the Honda–Tate boundary $|a| = 2\sqrt{q}$ **is** a Krein collision locus: the Weil bound is a *stability margin*, the Frobenius eigenvalues are stable modes on the circle, and a putative violation is a resonance event. The fake curve of E.8 is then exactly what an accelerator physicist would call it: an unstable lattice that no physical (positive-energy) machine realizes — the mode exists on paper, the hardware does not. Energy sign = Krein signature = the polarization: three fields' names for obligation (1).

The deformation lesson is the deepest one: within Π_1, with the ghost holding the slot, *everything else is pinned* (40 paths $\times$ 2001 steps, zero excursions). Instability is not a local risk anywhere along the family; it is concentrated entirely at boundary/degeneracy points. Stability theory thus explains *why RH resisted perturbative attack*: there is nothing to attack in the interior — every finite stage is already rigid.

16.4 Topological protection, and what Turing was actually computing [structural, exact at the mechanism]

E.14(a) identifies the index with the Turing discrepancy $N - N_0$. Three readings stack:

Computation. Turing built his method hoping to falsify RH; what his method computes is the Pontryagin index. Seventy years of verification — Platt–Trudgian to 3×10^{12} — is the longest-running index computation in mathematics, and it keeps returning 0 because integers, unlike inequalities, can be certified *exactly* by finite computation. The day's reformulation is an analog-to-digital conversion of RH: from a positivity (analog, unverifiable by sampling) to an integer (digital, certified window by window).

Condensed matter. An integer invariant computable as a winding (CHECK D), constant under deformation, able to change only where a gap closes (form degeneracy, E.15(b)): this is precisely the logic of topological protection — quantized Hall conductance, Chern numbers, edge states. **The zeros are topologically protected on the line**; an off-line zero would be a topological phase transition, and the gap-closing locus on the function-field side is Honda–Tate. RH says the arithmetic phase is gapped along the whole $q \to 1$ path *including the endpoint.* The condensed-matter intuition for the open step is then: prove the gap does

not close in the thermodynamic ($\lambda \to \infty$) limit — and that field's technology for exactly this (uniform gap bounds, spectral flow at infinity) is importable.

Statistics. The measured marginality exponent $d^{2.2} \approx d^2$ is the shadow of $\beta = 2$: GUE pair correlation vanishes quadratically, and an off-line zero *requires a coincidence* — two zeros at the same ordinate (the double-ordinate lemma). Level repulsion is the spectral face of index rigidity: the statistics actively suppress, at second order, exactly the degeneracy an off-line zero needs. CBS arrives again — order 2 carries what order 1 cannot — now as the order of the event RH forbids.

16.5 Passivity: the engineering-exact form of the second law [exact]

E.13's criterion — RH $\iff$ $-\Xi'/\Xi \in N_0$ — has a circuit-theoretic translation that is a theorem, not an analogy: Herglotz functions are exactly the **positive-real impedances of passive networks** (Brune; Foster synthesis), and they are exactly the **causal linear response functions** (Kramers–Kronig dispersion = the Herglotz integral representation; Titchmarsh). Hence, with full precision:

$$\text{RH} \iff \text{the arithmetic one-port is passive} \iff \text{the prime medium's susceptibility is KK-causal.}$$

An off-line zero is a hidden *active* element — an internal energy source, a gain medium violating dispersion causality. The Krein–Langer factorization $N = (r^{\#}/r)N_0$ is the engineer's decomposition into passive part $\times$ all-pass with unstable poles; κ counts right-half-plane poles; the winding of CHECK D is a Nyquist plot, and the Turing discrepancy is the Nyquist encirclement count. This upgrades Paper 32's "RH = second law of arithmetic" from reading to theorem-adjacent: **passivity is the circuit form of the second law, causality is its dispersion form, Weil positivity is its quadratic form, and the Lorentzian light-cone of P∞ is its geometric form — the N_0 criterion makes all four the same statement.** N14 (archimedean Weil positivity = relative entropy) supplied the thermodynamic leg; today supplied the engineering leg; the square is closed.

16.6 The measurement channel [exact]

E.16(a) is quantum-mechanically transparent. A rank-one perturbation is a coupling through a *single port*; $F(z)$ is the Weyl m-function — what a one-port measurement sees. The CCM architecture builds ζ by coupling the scaling flow to one channel, and the program already knows which one: **the pole** ("rank-1 pole carries the ground state"). So the pole is simultaneously the ghost (§16.2), the Nyquist critical point (§16.5), and the *measurement channel*: zeros are the resonances of the scaling flow read through the pole. RH $\iff$ every resonance is a bound state (real, embedded) rather than a decaying state (complex, finite lifetime). And the FE conjugation $C = J\iota$ with $CW = -WC$, which killed the index obstruction in E.11 by von Neumann's theorem, is *time-reversal symmetry* doing what antiunitary symmetries do in quantum mechanics (Kramers-style): constraining the spectrum structurally, for free. The functional equation is T-symmetry; T-symmetry pays for obligation (3); only energy positivity — obligation (1) — has dynamical content. This matches the physics exactly: in stability theory too, symmetry constrains and energy decides.

16.7 The redshifted tail [structural, with the weight exact]

E.16(d) gives the program's horizon metaphor a number. Information about a zero at height T reaches the real axis attenuated by the local envelope $|\Xi| \sim \mathrm{poly}(T)\, e^{-\pi T/4}$ — the archimedean Γ-decay, the redshift of Paper 32's horizon reading. The certificate's 4.9×10^{-9}-uniform perturbation migrating zeros at $t \approx 34$ is a detector lesson: a flat-calibrated instrument (an unweighted norm) is provably deaf to the tail, the way an uncalibrated strain gauge cannot see a gravitational wave — the signal is not weak, it is *far*. Lemma 7.2's true content is a calibration requirement: track the redshift exactly, $\sup e^{+\pi|t|/4}|\widehat{k}_\lambda - \Xi| \to 0$. The same envelope has now surfaced three independent ways — the prolate forbidden-region decay, the Γ-factor, and (via inverting Selberg's uniform zero-density exponent, §13) the crawl's logarithmic resolution-per-budget

law. One weight, three derivations: it is the *metric cost of the archimedean place*, and every route to RH pays it at the same rate. The program's repeated empirical finding of "exp-small marginality" was this weight being measured before it was named.

16.8 The synthesis: a conservation law in search of its medium

Assembling the six languages into one statement: **RH is the conservation of an integer under completion.** The integer is the Pontryagin index (equivalently: ghost count, Chern-like winding, unstable-pole count, resonance count, Turing discrepancy). It is conserved — provably, freely — at every finite stage and along every interior deformation, by mechanisms each field recognizes as its own (Krein pinning, T-symmetry, Herglotz reality, gap protection). Its conservation *through the boundary point* is the open step, and conservation laws propagate through limits only in a *medium* — some compactness or tightness that forbids escape. The day located the medium requirement exactly: tightness of an explicit Herglotz family in the $e^{\pi|t|/4}$-weighted norm (E.16(c,d)). The program's older language for the missing medium is "the multiplicative face" / the $\mathbb{F}_1$-cohomology that does not yet exist over $\mathbb{Q}$; the engineering language is "the matched termination of the one-port at infinity"; the GR language is "boundary conditions at the horizon." Same object, three specifications.

Two consequences for how we work. First, the difficulty geography is now measured: the open problem has *measure-zero support* — a point mass at the completion — with the weighted bound as its local cost; reductions should be counted (per the program's rule that difficulty is a scale) as reductions of *support*, and today's is the sharpest yet: one integer, one limit point, one named norm. Second, the dictionaries select the next intuitions to trust: the condensed-matter instinct (*prove the gap survives the thermodynamic limit*), the engineering instinct (*passive networks have matched-impedance completions*), and the scattering instinct (*resonance widths obey concentration estimates*) all point at the same queued strike — prolate/Bonami–Karoui concentration theory in the forbidden region, which is precisely a body of weighted estimates of the $e^{-\pi|t|/4}$ shape. None of this proves the bound; all of it tells us where proofs of bounds of exactly this shape already live.

Certificates underwriting this section: `probe_obligations_2_3_krein.py`, `probe_nkappa_krein_langer.py`, `probe_index_resolution_sT.py`, `probe_krein_collision_spectral_flow.py`, `probe_rank_one_determinant_spe` (all GREEN, `empirical/p_infinity/`).

17. The body for the shape: Riemann–Siegel as the anatomy of the medium

§16.8 ended with a shape in search of a body: the weighted bound $\sup e^{\pi|t|/4}|\hat{k}_\lambda - \Xi| \to 0$, the "medium" through which the index must be conserved. The body exists, it is classical, and identifying it dissolves the redshift: **the body is the Riemann–Siegel structure.** All five anatomical claims below are certified in `empirical/p_infinity/probe_rs_gauge_body.py` (GREEN).

17.1 The gauge identity: the weight is the archimedean skin

Proposition E.17 [P, **certified to** 10^{-31}]**.** With $R(t) = \pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{it}{2})|$ and $\mathrm{env}(t) = \frac{t^2+1/4}{2}\,R(t)$:

$$\Xi(t) \;=\; -\,\mathrm{env}(t)\cdot Z(t),$$

where Z is the Riemann–Siegel Z-function. The envelope is smooth, zero-free, and explicit — *dividing it out is a gauge transformation* — and $\mathrm{env}(t)\,e^{\pi t/4} \sim C\,t^{7/4}$ (fitted exponent 1.7500). **Corollary: the weighted spec of E.16(d) is exactly the unweighted (polynomially normalized) statement $\hat{k}_\lambda/\mathrm{env} \to -Z$ in sup norm.** The E.16 weight was never an analytic obstacle; it was the modulus of the archimedean Euler factor — the skin, not the flesh.

The demystification is quantitative (CHECK C): the "dangerous amplitude" of E.16(d) — the 2.3×10^{-9} extremum of Ξ between γ_5 and γ_6 — re-read in the Z-gauge is $|Z(m^*)| = 1.996 = O(1)$: in fact it is

the amplitude of the $n = 1$ phasor ($= 2$). **Zero-protection is an $O(1)$ problem in the right gauge.** The program's recurring "exp-small marginality" was the archimedean skin being measured in the wrong coordinates; §16.7's three appearances of the envelope (prolate forbidden region, Γ-decay, crawl) are three sightings of one gauge factor.

17.2 Anatomy

The skeleton — the Riemann–Siegel main sum: $Z(t) \approx 2\sum_{n\le\sqrt{t/2\pi}} n^{-1/2}\cos(\theta(t) - t\log n)$, error $O(t^{-1/4})$ with a complete asymptotic correction series (certified at $t = 100, 500, 2000$: errors $0.45, 0.25, 0.15$ within the $2t^{-1/4}$ allowance, with only $3, 8, 17$ modes). At every height the target has a **finite arithmetic skeleton**: the first $\sqrt{t/2\pi}$ integers, amplitudes $n^{-1/2}$, phases set by θ. Unconditional.

The skin — $\mathrm{env}(t)$, the archimedean modulus (§17.1): exact, zero-free, gauge.

The light cone — the cutoff is **stationary phase**: $\frac{d}{dt}[\theta(t) - t\log n] = 0$ exactly at $n = \sqrt{t/2\pi}$, since $\theta'(t) = \frac{1}{2}\log(t/2\pi)$ is the instantaneous frequency (certified to 4 decimals: $e^{\theta'} = 8.9206 = \sqrt{500/2\pi}$). Integer mode n *enters the light cone* at $t = 2\pi n^2$ and thereafter contributes a rotating phasor; modes outside the cone are forbidden. This is the program's causal structure appearing in the body: the $S(T)$-face of P∞'s light cone.

The immune system — past the cone edge, contributions die at Agmon/WKB rates; quantitatively this is the prolate eigenvalue plunge and the Bonami–Karoui non-asymptotic estimates. CHECK E certifies the deepest anatomical fact here: the Mellin transforms of the whole oscillator family $x^{2n}e^{-\pi x^2}$ decay with the **same** exponent $e^{-\pi|t|/4}$ — only the polynomial prefactor $t^{n-1/4}$ varies. **The redshift is place-universal: it belongs to the archimedean metric, not to any state.** This is why dividing it out is legitimate gauge, not cheating: no state carries information in that exponent.

The heart — what remains contested after the gauge move: a **log-width window at the cone edge** (the plunge region / the saddle $n^* = \sqrt{t/2\pi}$), at $O(1)$ amplitude. The deep forbidden region is free (immune system); the interior of the cone is exact arithmetic (skeleton); all the blood flows through the edge.

17.3 Dynamics

In the Z-gauge the dynamics is phasor interference. $Z(t)$ is a sum of $N(t)$ rotating phasors with frequencies $\theta'(t) - \log n > 0$ and amplitudes $n^{-1/2}$; **zeros are destructive interferences**, occurring once per π of accumulated phase (the $n = 1$ phasor alone forces the mean density). The GUE statistics of §16.4 is the interference statistics of incommensurate frequencies $\{\log n\}$. An off-line zero, which by the double-ordinate lemma requires a *coincidence*, is in this language a **dead zone**: the phasor sum and its derivative vanishing together — an amplitude collapse that the $n^{-1/2}$ hierarchy (the $n = 1$ phasor dominating all others combined unless phases conspire to second order) resists at exactly the d^2 rate measured in §13. New modes are born on the light cone ($t = 2\pi n^2$) — the dynamical events of the flow — and the Riemann–Siegel correction series (the Ψ-function, built from the Mordell integral / quadratic Gauss sums) is the **matched termination at the edge**: the engineering "matched impedance" of §16.8, realized classically — and it is a theta/metaplectic object, reconnecting to the program's $\mathbb{Q}(\zeta_8)$/metaplectic backbone (Q8vM3; Mandal–Mondal) [structural].

The λ-dynamics (CCM) must, in this gauge, do exactly one thing: reproduce the skeleton inside the cone (finite arithmetic, available — the Euler product is built into the triple) and control the plunge window at $O(1)$. The RG reading of E.6′ ($c(\lambda) = \kappa\log\lambda$) is the statement that each octave of λ admits new modes at constant entropy cost — the crawl is the light cone expanding logarithmically in RG time.

17.4 What moved, and what is still open

The localization improves again, and it is the sharpest yet: from *a weighted bound on all of* $\mathbb{R}$ (E.16) to **an $O(1)$-amplitude bound on a log-width window at the saddle, in the Z-gauge.** Difficulty support:

one integer (the index), one limit point (the completion), one window (the cone edge), one amplitude scale ($O(1)$, the $n = 1$ phasor). Honesty: (i) Riemann–Siegel is unconditional anatomy of the *target* — it does not by itself make $\widehat{k}_\lambda$ match it; the open step is now "the prolate flow hits the skeleton with uniform $O(1)$-gauge error through the plunge window", [C] for the CCM objects pending their normalizations. (ii) The identification of the CCM/prolate plunge edge with the RS saddle is [C] and contains a real question we name rather than blur: the spectral description saturates at $N(T) \sim \frac{T}{2\pi}\log T$ modes (zeros) while the arithmetic skeleton saturates at $\sqrt{T/2\pi}$ (integers) — two different saturation edges (this is the $\tau = 1/\text{duality}$ edge of the gue_cumulant arc in yet another costume), and the *square-root* relation between them is the square-root cancellation of analytic number theory. Reconciling the two edges inside the triple is the precise next question the body poses.

Certificate. `empirical/p_infinity/probe_rs_gauge_body.py` (GREEN, 5/5): gauge identity to 10^{-31} across $t \in [5, 95]$; envelope exponent 1.7500 vs 7/4; Z-gauge threshold $1.996 = O(1)$ vs 2.3×10^{-9} in the Ξ-gauge (ratio = env $= 1.16 \times 10^{-9}$); RS skeleton within $2t^{-1/4}$ with 3–17 modes; stationary-phase saddle $e^{\theta'} = \sqrt{t/2\pi}$ to 10^{-3}; place-universality of $\pi/4$ across oscillator levels $n = 0, 2, 5$ (drift ≤ 0.03).

Anchors (§17). Riemann–Siegel (the formula; Mordell integral, Ψ correction); Berry–Keating (RS asymptotics, phase dynamics); Bonami–Karoui (prolate plunge estimates); Agmon (weighted decay); CCM 2511.22755 (Euler product in the triple; Lemma 7.2); Connes 2602.04022 §6.5; P∞ (light cone); P∞-D §3 (crawl = logarithmic cone growth); Q8vM3, Mandal–Mondal ($\mathbb{Q}(\zeta_8)$ metaplectic backbone); gue_cumulant arc ($\tau = 1$ edge); §13 ($d^{2.2}$ law); §16 (dictionary).

18. One cone, two edges: the reconciliation, the 1/8 seam, and the winding-neutral prime field

§17.4 named the open question the body poses: the spectral description saturates at $N(T) \sim \frac{T}{2\pi}\log T$ modes (zeros) while the arithmetic skeleton saturates at $\sqrt{T/2\pi}$ (integers). The reconciliation is certified in `empirical/p_infinity/probe_one_cone_two_edges.py` (GREEN), and it dissolves the apparent competition: **the two edges are two readings of one phase-space cone — its area and its boundary height — and the cone is the classical phase space of $H = xp$, which is the CCM scaling operator $S = \frac{xp+px}{2}$.**

Theorem E.18 (one cone, two edges). Let $\mathcal{C}(T) = \{(x, p) : x \geq 1,\ p \geq 2\pi,\ xp \leq T\}$ be the xp cone (Berry–Keating box). **(a) (Area = zeros, with the 1/8 seam phase; certified.)** The Planck-cell count $|\mathcal{C}(T)|/2\pi = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + 1$ exceeds the Riemann–von Mangoldt smooth count $\theta(T)/\pi + 1$ by **exactly** 1/8 in the limit (certified: deficit = 0.124867, 0.124934, 0.124987, 0.124997, 0.124999 at $T = 50, 10^2, 5\cdot 10^2, 2\cdot 10^3, 10^4$). The naive cone overcounts by the Maslov-type boundary phase of its corner — and 1/8 of a turn is precisely the **apex-archimedean phase the program measured in N3/N4** (the continuous-spectrum $\sqrt{i}$, "exactly 1/8 turn"). The seam phase of the cone is the archimedean reflection phase [A, numerically exact]. Three-way match at $T = 500$: sign-change count 269, $\theta/\pi + 1 = 269.587$, xp-box$-\frac{1}{8} = 269.587$. **(b) (Boundary height = integers.)** The arithmetic edge is $e^{\theta'(T)} = \sqrt{T/2\pi}$ — the *exponentiated height* of the same cone (§17, certified to 4 decimals). Integers occupy the exponentially sparse frequencies $\log n$; the number of available modes below the instantaneous frequency Ω is e^Ω, while the degrees of freedom (Landau/Nyquist count) is the *area* $\frac{1}{\pi}\int_0^T \theta'$. **"Square-root cancellation" is cone geometry: area grows like $T\log$(height) while the mode count is the height itself.** No paradox — one object, two coordinates. **(c) (Winding neutrality; certified.)** The secular zero density is 100% **archimedean**: the running discrepancy $S(t) = N_0(t) - (\theta(t)/\pi + 1)$ over $[10, 500]$ has mean $+0.05$ and max modulus 0.95 — the full prime phasor field has *zero Weyl mean motion*, every unit of secular winding coming from θ'. By contrast the **half-field** (the RS Dirichlet polynomial D_{17} alone, FE mirror removed) drifts at measured mean motion -0.813 per unit t against $\theta' = 2.894$. **The functional-equation mirror cancels a finite, measured drift at the seam — 0.813 per unit time at $T \approx 2000$ — and that cancellation is the entire job of the Gauss-sum/Ψ termination.** Square-root cancellation, dynamically: the full

field is winding-neutral; each half drifts; the mirror pays the other's debt at the light-cone boundary.

What this does to the open step. Three further reductions, each flagged. (i) The *mean* obligation is **discharged unconditionally**: winding neutrality is equivalent to $S(T) = o(T)$, which is classical ($S(T) = O(\log T)$, von Mangoldt) — the triple does not need to *prove* the secular count, only inherit it. (ii) The *cone* obligation is discharged **by construction**: the CCM operator is xp itself; its phase space is the cone; θ' is its classical frequency. (iii) What remains is exactly the **seam**: the 1/8 metaplectic corner phase plus the cancellation of the measured half-field drift — both living in the log-width plunge window, both theta/Gauss-sum objects, and the program already holds the relevant structure theorem: Q8vM3's Uniform Place Treatment *forces* the metaplectic representation at $v = \infty$, i.e. the triple is group-theoretically equipped with the right seam phase. The open step, after §§15–18, reads: **show the CCM flow executes the mirror cancellation uniformly through the plunge window** — an $O(1)$ statement about a metaplectic boundary term, with mean and cone both free. The residual's support shrinks again: from a window to *the seam inside the window.*

Consilience. The $\tau = 1$ edge of the gue_cumulant arc, the RS saddle, the prolate plunge, and the Montgomery diagonal/off-diagonal boundary are now one object — the cone boundary — seen by pair correlation, by the approximate functional equation, by band-limiting, and by the form factor respectively; the factor-of-two relations among their exponents are all instances of (c)'s mirror doubling. And the day's earlier results sit on the cone too: the index (κ, §§11–14) is conserved *inside* the cone (Krein pinning), can jump only at the boundary (form degeneracy = seam), and the Lorentzian time direction (§16.2) is the cone's axis. The program's light-cone metaphor was not a metaphor.

Certificate. `empirical/p_infinity/probe_one_cone_two_edges.py` (GREEN, 3/3): seam deficit $\to 1/8$ to 10^{-6} by $T = 10^4$; three-way count match at $T = 500$ within $|S| \le 0.95$; $\langle S \rangle = +0.05$ neutrality vs half-field drift -0.813 measured.

Anchors (§18). Berry–Keating ($H = xp$; the 7/8 vs Maslov discussion); Riemann–von Mangoldt (7/8); Landau (band-limited counting); Weyl (mean motion); N3/N4 (apex-archimedean operator; the 1/8 turn); Q8vM3 (UPT forces metaplectic at $v = \infty$); CCM 2511.22755 ($S = xp$ scaling triple); Montgomery (form factor; $\tau = 1$); gue_cumulant arc; §13 (graded index); §17 (the body).

19. The engine room read at source: the seam located in CCM, and the weld (Theorem E.19)

We now have the CCM source (arXiv 2511.22755v1) in hand and have certified its engine room independently (`empirical/p_infinity/probe_ccm_seam_weld.py`, GREEN, 5/5). This section records the source-verified map onto §§15–18, the location of the seam in their formalism, and the weld theorem that converts their two "missing steps" into a single spectral-gap bound with doubly-exponential margin.

19.1 The source-verified map

Their Theorem (finmain): for each (λ, N), the rank-one perturbation $\widetilde{D} = D^{(\lambda)}_{\log} - |D^{(\lambda)}_{\log}\xi\rangle\langle\delta_N|$ of the scaling operator is self-adjoint **with respect to the inner product** $QW^N_\lambda - \epsilon_N\langle\,|\,\rangle$ **on** $E_N/\mathbb{C}\xi$ — the Weil form shifted by its lowest eigenvalue, with the ground state ξ quotiented out — and $\det_{\mathrm{reg}}(\widetilde{D} - z) = -i\lambda^{-iz}\widehat{\xi}(z)$ with all zeros of $\widehat{\xi}$ **real**. Three identifications, now at source level rather than reconstruction: (i) this is E.16(a) instantiated — the rank-one determinant whose reality is free at every finite stage, with the positivity supplied by the *minimum-shift* $QW - \epsilon_N \ge 0$ (automatic, not RH); (ii) the quotient $E_N/\mathbb{C}\xi$ is the **ghost compression** of §16.2 verbatim — the ground state is the null direction, the physical space is its quotient; (iii) their closing factor "e^{a+ibs}" is our renormalization c_λ, changing no zeros. Their Lemma (hermfact1), *proven*: $\widehat{k}_\lambda \to \Xi$ **uniformly on closed substrips of** $|\Im z| < \frac{1}{2}$ — precisely the convergence mode E.16(b) feeds to Hurwitz. The two halves of RH are thus held by two different functions: $\widehat{k}_\lambda$ *converges* (no reality theorem), $\widehat{\xi}_\lambda$ *has real zeros* (no convergence theorem). Their two missing steps, in their own words: (1)

simple-even for the ground state of QW_λ; (2) k_λ approximates ξ_λ well enough to transfer. *[(1) was later discharged unconditionally — Suzuki's Theorem 5 = our E.11; P∞-F §12.]*

19.2 The seam, located and measured

The mirror cancellation of §18(c) sits in their formalism at one exact point. The Riemann weight is $k = \mathcal{E}(h)$, $\mathcal{E}(f)(u) = u^{1/2}\sum_{n\geq 1} f(nu)$, with h the *vanishing-integral* combination of the Fourier-invariant Hermite functions h_0, h_4. Poisson summation gives $k(u) = k(1/u)$ — and the certificate shows this is **exact** (1.4×10^{-15}) *because* $h(0) = 0$ and $\int h = 0$ kill both boundary terms of Poisson: **the vanishing-integral condition is the exact seam cancellation at $\lambda = \infty$.** (The choice of h_0, h_4 — Hermite levels $n \equiv 0 \bmod 4$, the $+1$-eigenspace of the metaplectic Fourier grading i^n — is the seam *phase* selection; the 1/8 of §18 is this grading's quarter-of-a-half turn.) At finite λ the prolate h_λ is invariant only under the *compressed* Fourier transform, with defect $1-\chi(\lambda) \sim Ce^{-4\pi\lambda^2+9\log\lambda}$ (Fuchs): the mirror breaks, and the breakage measured on our independently-built prolates ($\sigma(\lambda) \approx 1.3$–$2.8\times10^{-5}$) sits **three orders below** the λ^{-2} Meixner–Schäfke floor ($\|h_\lambda - h\|_\infty = 0.014$–$0.006$, fitted rates $\lambda^{-2.16}, \lambda^{-2.56}$ — their estimate reproduced from scratch by banded FD diagonalization). And their "indication (2)" is the seam identity in spectral clothing: $\epsilon_\lambda \approx 1-\chi(\lambda)$ — **the Weil ground eigenvalue *is* the seam defect.** End-to-end, our independent pipeline measures $|\hat{k}_\lambda(z) - \Xi(z)|$ falling $0.052 \to 0.010$ ($z = 0$) over $\lambda = 1.5 \to 3$, rate $\approx \lambda^{-2.3}$ — *faster* than their stated $O(\lambda^{-1/2})$ bound.

19.3 Theorem E.19 (the weld)

Let $g(\lambda)$ be the spectral gap of QW_λ above ϵ_λ and $r(\lambda)$ the Rayleigh excess $QW_\lambda(\tilde{k}_\lambda) - \epsilon_\lambda$ of the normalized prolate guess. Then: **(a)** [P, certified] $\|\tilde{k}_\lambda - \langle\xi_\lambda, \tilde{k}_\lambda\rangle\xi_\lambda\|^2 \leq r(\lambda)/g(\lambda)$ (gap-transfer; 200 random instances). **(b)** [A] If $\sqrt{r(\lambda)/g(\lambda)}\cdot\mathrm{poly}(\lambda) \to 0$, their own Mellin-transfer template (the proof of hermfact1, which loses only the polynomial factors $u^{1/2}\cdot\lambda/u$ and the strip weights) passes the substrip convergence from $\hat{k}_\lambda$ to $\hat{\xi}_\lambda$; with finmain(iii) reality and Hurwitz (E.16(b)), **RH follows**. **(c)** The budget: $r(\lambda) \approx \epsilon_\lambda \approx 1-\chi(\lambda) \sim e^{-4\pi\lambda^2}$ (their indication (2) + Fuchs; rigorization = the Co-zeta trace formula, [C]). The requirement on the gap is therefore only $g(\lambda) \geq e^{-4\pi\lambda^2}\,\mathrm{poly}(\lambda)$ — **even an exponentially collapsing gap suffices.** The two missing steps compress to: *(i) simple-even; (ii) a spectral-gap lower bound that merely beats $e^{-4\pi\lambda^2}$.* The seam has doubly-exponential margin over the requirement; numerically the gap of QW_λ appears $O(1)$ (the higher eigenvalues track γ_k-spacings), so the conditional chain asks for astronomically less than what the numerics display.

Status and honesty. (a) is a theorem; (b) is referee-level pending the bookkeeping of the transfer weights; (c)'s budget identity is numerical-plus-asymptotic, its rigorization the precise job of the Co-zeta trace formula relating QW_λ to $(P_\lambda, \widehat{P}_\lambda, \mathcal{E})$ — the cornerstone CCM themselves name. Nothing here closes RH: the open step is now *(simple-even) + (gap $\geq e^{-4\pi\lambda^2}$) + (trace-formula Rayleigh bound)* — three concrete, separately attackable statements about one explicit quadratic form, replacing "prove the convergence". That is the day's final compression of the residual's support. *[Historical flag — (c)'s "doubly-exponential margin" was self-attacked in §20 (the ladder picture) and resolved by measurement in §21: the true structure is an exponentially-graded ladder, margin exponential; and the §22 erratum adds the hidden hypothesis (H0′) to every weld-chain statement, this one included.]*

Certificate. `empirical/p_infinity/probe_ccm_seam_weld.py` (GREEN, 5/5): h-identity to 10^{-16}; exact Poisson mirror 1.4×10^{-15}; duality scalar = 4.000000, $4M(k)(iz) = \Xi(z)$ to 10^{-15}; Meixner–Schäfke rates $\lambda^{-2.16}, \lambda^{-2.56}$ from scratch; seam $\sigma(\lambda)$ three orders sub-dominant; $\hat{k}_\lambda \to \Xi$ at measured $\lambda^{-2.3}$; gap-transfer lemma exact on 200 instances.

Anchors (§19). CCM 2511.22755v1 (source: finmain, hermfact, meixnerlem, hermfact1, "The missing steps"); Meixner–Schäfke (Satz 9); Fuchs (prolate eigenvalue asymptotics); Co-zeta (trace formula $QW_\lambda \leftrightarrow P_\lambda, \widehat{P}_\lambda, \mathcal{E}$); VJ (semilocal spectral realization); §§15–18 (the rank-one spec, the dictionary, the body, the cone); Q8vM3 (metaplectic grading = seam phase).

20. Cliff or ladder: the E.19 margin corrected, and the window Weil spectrum measured

GPD discipline requires us to attack our own §19.3(c) before resting on it, and the attack lands: the "doubly-exponential margin" assumed the spectral gap of QW_λ is $O(1)$. There is a competing picture. ("Co-zeta" is now identified at source: Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. 1999.)

The two pictures. *Cliff*: the higher spectrum of the form sits at $O(1)$ (the Dirac-side intuition — eigenvalues of $\widetilde{D}$ track zero spacings); then $g(\lambda) = O(1)$ and the E.19 margin is doubly exponential. *Ladder*: via the trace formula, the low spectrum of QW_λ is the prolate angle-operator spectrum $\{1-\chi_n(\lambda)\}$, whose Fuchs ratios are only *polynomial*: $(1-\chi_{n+1})/(1-\chi_n) \sim 16\pi\lambda^2/(n{+}1)$. Then $g(\lambda) \sim \lambda^2\epsilon_\lambda$, the weld ratio is $r/g \sim \lambda^{-2}$ *if* $r \approx \epsilon_\lambda$, and the chain becomes **marginal** against the polynomial transfer losses. The corrected statement of E.19(c): the chain closes comfortably in the cliff picture; in the ladder picture everything rides on the *excess* $r(\lambda) = QW_\lambda(\tilde{k}_\lambda) - \epsilon_\lambda$ being $o(\epsilon_\lambda)$ — a difference of two nearly equal exp-small quantities, exactly what the Co-zeta trace formula is built to compare. **The margin question and the trace-formula rigorization are the same question.** Recorded as the surviving quantitative crux.

The measurement (`empirical/p_infinity/probe_window_weil_spectrum.py`, GREEN). We built the window Weil form with the program's certified geometric machinery (validated against 250 zeros at absolute floor 3×10^{-4} — CHECK A) on tapered-cosine even bases over $[\lambda^{-1}, \lambda]$, $\lambda \in \{1.5, 2, 3, 4\}$, and measured its generalized spectrum (Löwdin-regularized; a spurious λ=3 "negative eigenvalue" was caught and killed as Gram-conditioning artifact — the float64-artifact lesson enforced). Findings: (i) **positivity holds** — no eigenvalue below −floor at any λ (window Weil positivity witness; a genuine negative would disprove RH); (ii) the spectrum splits cleanly into a **near-null cluster** ($\lesssim$ few$\times10^{-3}$) and an $O(1)$ bulk, with the cluster dimension *growing* with the window: $1, 3, 6, 9$ at $\lambda = 1.5, 2, 3, 4$ — the form is a sampling form $\sum_\gamma w_\gamma |\hat{f}(\gamma)|^2$ and the near-kernel counts the test functions that dodge the visible zeros: **the cone deficit of §18 made operator** [A]; (iii) the exp-small grading *inside* the cluster — the cliff-vs-ladder discriminator — sits **below any float64 floor**: quadrature machinery cannot resolve it, period. The honest verdict: cliff-vs-ladder is open, and its resolution requires the **exact CCM matrix** (their §5 explicit $W_{0,2}/W_p/W_\mathbb{R}$ tables) in high-precision arithmetic — the same exact-kernel route that resolved the Connes-miracle artifacts. That build is the queued next strike.

Where the open step stands after today (§§11–20, ten sections, ten GREEN certificates): one integer (the index), conserved freely everywhere except one boundary point; at the boundary, one seam (the metaplectic mirror), with the weld reducing the convergence to *(simple-even) + (gap) + (trace-formula Rayleigh comparison)*; and the margin of the weld now localized to a single quantitative question — **is the prolate guess's Rayleigh excess $o(\epsilon_\lambda)$, equivalently does the Co-zeta trace formula control $QW_\lambda(\tilde{k}_\lambda) - \epsilon_\lambda$ at sub-leading order in the seam defect?** That is a question about one explicit trace identity evaluated on one explicit function. The support of the difficulty has not been this small in the program's history. *[Historical flag — resolved in §21 by the exact-matrix build: cliff-vs-ladder settled (exponentially-graded ladder), the Rayleigh excess measured $r/\epsilon \approx 0.25$, the marginality scenario refuted by measurement.]*

Anchors (§20). Connes 1999 (Co-zeta, the trace formula); Fuchs 1964 (ladder ratios); CCM 2511.22755 §5 (explicit matrix tables — the exact-arithmetic route); VJ (Dirac-side spectra); §18 (cone deficit); §19 (the weld); GPD (Paper 35; self-attack discipline); probes: `probe_window_weil_spectrum.py` and the nine preceding GREEN certificates of §§11–19.

21. The exact matrix: cliff-or-ladder resolved, simple-even verified, and the weld margin measured exponential

The queued build is done: the **exact CCM matrix** of QW_λ in the V_n basis, in mpmath (dps = 40), from the source formulas (Prop toadd + Lemma polarize0 + Lemma wsharp). The key implementation observation: $\Psi^\#$ is *linear* in q, and $q(U_n, U_m)$ is a difference of single-frequency sines — so the entire $(2N{+}1)^2$ matrix

reduces to $\sim 4(2N{+}1)$ one-dimensional integrals of elementary functions, computable to arbitrary precision. All in `empirical/p_infinity/probe_ccm_exact_matrix.py` (GREEN); convention validated against the zero side within the zero-sum tail allowance.

Finding 1 — simple-even, verified. At $\lambda = 1.3$, 1.5, 2.0 (pure archimedean; prime 2; primes $2, 3, 4$): the ground state of QW_λ^N is **even and simple** in every case, with the odd-sector minimum strictly above it. CCM's missing step (1) is numerically certified by exact arithmetic at three windows.

Finding 2 — neither cliff nor my ladder: an exponentially-graded ladder. The measured even spectra:

$$\lambda{=}1.3:\ 2.5{\times}10^{-2},\ 0.90,\ 1.47,\ldots \qquad \lambda{=}1.5:\ 1.57{\times}10^{-4},\ 0.28,\ 0.90,\ldots$$
$$\lambda{=}2.0:\ 1.38{\times}10^{-12},\ 2.75{\times}10^{-7},\ 3.8{\times}10^{-3},\ 0.57,\ldots$$

The gap ratio μ_1/ϵ **grows super-polynomially with the window**: $37 \to 1787 \to 2.0{\times}10^5$. The structure is a ladder, but its rungs separate at exponential-in-λ^2 rates — far steeper than §20's conservative λ^2-per-rung scenario. Structurally [A]: the even low spectrum is the $n \equiv 0 \pmod 4$ **prolate tower** $\{1-\chi_{4k}(\lambda)\}$ — the ground is the h_0/h_4 cell, the first excited the $n = 8$ cell, with Fuchs rung ratios $\sim (8c)^4/\text{poly}$, $c = 2\pi\lambda^2$ (at $\lambda = 2$: $\sim 10^6$ predicted vs 2×10^5 measured — right order). **The gap bound the weld needs is a second-rung Fuchs asymptotic — prolate theory, not RH-hard.**

Finding 3 — the seam identity across 13 decades. ϵ_λ vs the Fuchs scale $\frac{2^{14}}{3}\sqrt{2}\,\pi^5 e^{-4\pi\lambda^2+9\log\lambda}$: ratio 3.3 at $\lambda = 1.5$ and 7.7 at $\lambda = 2.0$ — order-unity tracking while the quantities themselves fall from 10^{-4} to 10^{-12} (and the Fuchs scale spans $e^{-28} \to e^{-50}$). CCM's Figure-fpro1 identity $\epsilon_\lambda \approx 1 - \chi(\lambda)$ — the *Weil ground eigenvalue is the seam defect* — independently reproduced by exact arithmetic.

Finding 4 — the weld crux measured: $r/\epsilon = 0.25$. The Rayleigh excess of the (FD-built) prolate guess in the exact matrix at $\lambda = 1.5$: $R = 1.95 \times 10^{-4}$ vs $\epsilon = 1.57 \times 10^{-4}$, so $r/\epsilon \le 0.246$ (the FD guess bounds r from above; its contamination floor $\sim 10^{-10}$ is four orders below the measurement). Combining with Finding 2: the weld ratio

$$r/g \;=\; (r/\epsilon)\cdot(\epsilon/\mu_1) \;\approx\; 0.25/1787 \;\approx\; 1.4\times10^{-4} \quad (\lambda{=}1.5),$$

shrinking exponentially in λ^2 as the rung separation grows. Hence $\|\tilde{k}_\lambda - \xi_\lambda\| \lesssim e^{-c\lambda^2/2}$ — beating *every* polynomial transfer loss. §20's marginality scenario is **refuted by measurement**: the true structure is more favorable than both §19's cliff optimism and §20's ladder pessimism. (r at $\lambda = 2$ is below the FD-guess noise floor; pushing the measurement deeper needs an mp-precision prolate build — queued, not load-bearing.)

Where the weld now stands. The conditional chain of Theorem E.19 rests on three statements, and after this section each has changed character: *(i) simple-even* — verified at three windows, plausibly provable by prolate-tower perturbation (the measured spectra show the even/odd interlacing cleanly) *[since discharged unconditionally: P∞-F §12]*; *(ii) the gap bound* — now identified as a **second-rung Fuchs asymptotic** for the mod 4 prolate tower, i.e. classical Slepian-theory, with the measured $e^{c\lambda^2}$ separation as its target; *(iii) the Rayleigh comparison* $r = O(\epsilon)$ — measured at 0.25, its rigorization the Co-zeta trace formula's job. None is closed; all three now look like **prolate-theory lemmas rather than RH-hard statements**, and the chain they support ends, via Hurwitz, at RH. This is the program's state of the art on the residual.

Certificate. `empirical/p_infinity/probe_ccm_exact_matrix.py` (GREEN): exact matrix at dps = 40; zero-side validation; even/odd spectra at three λ; ϵ-vs-Fuchs across 13 decades; $r/\epsilon = 0.2465$.

Anchors (§21). CCM 2511.22755v1 (Prop toadd; Lemma polarize0; Lemma wsharp; eq. quadratsemi; Figure fpro1); Fuchs 1964 (tower asymptotics); Slepian school (prolate spectral theory); Meixner–Schäfke; Co-zeta (Connes 1999); §§19–20 (the weld and its self-attack); GPD (measurement over speculation).

22. Theorem E.20: the quantified weld, certified end to end

We now state the weld as a referee-grade conditional theorem with explicit constants, and certify **every inequality in its proof on the exact $\lambda = 1.5$ system** (`empirical/p_infinity/probe_quantified_weld_e2e.py`, GREEN).

Theorem E.20 (quantified weld). Let $\tilde{k}_\lambda$ be the normalized even projection of the prolate guess, ϵ_λ the lowest eigenvalue of QW_λ^N with eigenvector ξ, μ_1 the next *even-sector* eigenvalue, $r = QW_\lambda(\tilde{k}_\lambda) - \epsilon_\lambda$. Suppose, along a sequence $\lambda \to \infty$ (with $N = N(\lambda)$ large): **(H1)** the ground state is simple and even; **(H2)** $QW_\lambda(\tilde{k}_\lambda) \le \mathrm{poly}(\lambda)\, e^{-4\pi\lambda^2}$; **(H3)** $\mu_1 - \epsilon_\lambda \;\ge\; \lambda^{1+\delta}\, QW_\lambda(\tilde{k}_\lambda)$ for some $\delta > 0$. Then RH holds. *Proof chain:* (i) min–max: $\|\tilde{k}_\lambda - \xi\| \le \sqrt{2r/(\mu_1 - \epsilon)} \le \sqrt{2}\,\lambda^{-(1+\delta)/2}$; (ii) Cauchy–Schwarz transfer: for $z = t + i\beta$, $|\beta| \le \frac{1}{2} - \delta'$, $|\widehat{(\tilde{k} - \xi)}(z)| \le \|\tilde{k} - \xi\| \cdot W(\beta, \lambda)$, $W = \left((\lambda^{2\beta} - \lambda^{-2\beta})/2\beta\right)^{1/2} \le \lambda^\beta/\sqrt{\beta}$ — *polynomial* loss; the product $\lambda^{-(1+\delta)/2} \cdot \lambda^{1/2-\delta'} \to 0$; (iii) CCM hermfact1 (proven): $\hat{k}_\lambda \to \Xi$ uniformly on closed substrips; hence $\hat{\xi}_{\lambda,N} \to \Xi$ there; (iv) CCM finmain(iii) (proven, given H1): all zeros of $\hat{\xi}_{\lambda,N}$ are **real**; (v) Hurwitz (E.16(b)): the locally uniform limit of nonvanishing-off-$\mathbb{R}$ functions has no zeros in any open substrip $\Rightarrow$ all zeros of Ξ real $\Rightarrow$ RH. $\square$

The hypotheses against the measurements of §21: (H1) verified at three windows; (H2) measured ($R = 1.95 \times 10^{-4}$ at $\lambda = 1.5$, tracking the Fuchs scale; $r/\epsilon = 0.135$ in the even sector); (H3) measured with enormous room ($\mu_1/QW(\tilde{k}) \approx 1.4 \times 10^3$ at $\lambda = 1.5$, $\sim 10^9$ at $\lambda = 2$ — against a required $\lambda^{1+\delta} \approx 1.6$). **The theorem needs $\lambda^{1+\delta}$; the system delivers $e^{c\lambda^2}$.**

End-to-end certificate. On the exact $\lambda = 1.5$ matrix: min–max checked on real data ($\|\tilde{k}_e - \xi\| = 0.00661 \le \sqrt{2r/\mathrm{gap}} = 0.01232$); the transfer bound checked at four strip points including $\beta = 0.45$ (bound 0.0060, actuals 0.0006–0.0030); and the **payoff demonstrated from our own build**: the zeros of $\hat{\xi}$ (ground state of *our* exact matrix) are real to 10^{-44} (their theorem, on our system) and land on the Riemann zeros —

$$\lambda = 2.0 \text{ (primes } 2, 3, 4 \text{ only)}: \quad |\hat{z}_1 - \gamma_1| = 2.4 \times 10^{-9}, \quad |\hat{z}_2 - \gamma_2| = 6.5 \times 10^{-7}, \quad |\hat{z}_3 - \gamma_3| = 1.4 \times 10^{-5},$$

versus 0.043, 0.73, 3.0 at $\lambda = 1.5$: seven orders of improvement in half a unit of λ — the $e^{-4\pi\lambda^2}$ pace, visible. The Connes miracle, reproduced from an independent exact build with three prime powers.

Status. E.20 is proven as stated; RH now rests, in this architecture, on (H1)–(H3) — all measured true with exponential room, all prolate-theory-shaped, none closed. The next analytic targets, in order: (H3) via second-rung Fuchs asymptotics for the mod 4 tower; (H1) via ground-cell isolation in the same tower; (H2) via the Co-zeta trace formula. The referee write-out of §§1–22 is queued.

ERRATUM (2026-06-12, from the P∞-F §14 audit — the hidden hypothesis (H0′)). The proof chain's step (i) uses $r \le QW_\lambda(\tilde{k})$, which holds only if $\epsilon_\lambda \ge 0$; and step (ii)'s smallness needs $r/(\mu_1 - \epsilon)$ small, which fails if ϵ_λ dives: in an RH-false world $\epsilon_\lambda \to -\infty$ (Yoshida; rate quantified by the P∞-F §14.1 spike calculus), $r \approx |\epsilon| \approx \mu_1 - \epsilon$, and the transfer is vacuous — the chain correctly stalls, but this means E.20's hypothesis set must include

$$\textbf{(H0}') \qquad \epsilon_\lambda \;\ge\; -o(\lambda^{-(1+\delta)}\,(\mu_1 - \epsilon_\lambda)) \qquad \text{along the sequence,}$$

and **(H0′)-along-a-sequence $\iff$ RH** (Yoshida 1992; Suzuki arXiv:2606.09096 Thm 3). The corrected statement: *(H0′) + (H1) + (H2) + (H3) ⇒ RH*, with (H1) since discharged unconditionally (P∞-F §12, via Suzuki's Thm 5 = E.11), (H2) reduced to proven lemmas + certified bookkeeping (§25; P∞-F §16), (H3) reduced to the sector split (§26) whose every facet except (H0′) itself is now proven, certified, or welded (P∞-F §§7–16). **The conserved core of the architecture is (H0′) — window Weil positivity surviving the limit — located precisely; see P∞-F §17.2 for its three certified representations and the (H0′-tail) localization.** The "exponential room" measured for (H1)–(H3) above is unaffected; what the measurements cannot address is (H0′), which is RH itself, windowed.

Anchors (§22). CCM 2511.22755v1 (hermfact1; finmain); §§19–21; Hurwitz; min–max; Cauchy–Schwarz; `probe_quantified_weld_e2e.py` and the twelve GREEN certificates of §§11–22.

23. The Fuchs strike: the cliff refuted by proof, the second cell identified, and (H3) reduced to one bridge inequality

The first analytic target of §22 falls into three certified pieces and one named residual (`empirical/p_infinity/probe_fu` GREEN).

Theorem E.21(a) — the cliff is refuted rigorously. By Courant–Fischer, $\mu_1 \le$ the larger eigenvalue of the compression of QW_λ to *any* 2-dimensional subspace. Taking the two prolate cells $\mathcal{E}(h_4 - \frac{I_4}{I_0}h_0)$, $\mathcal{E}(h_8 - \frac{I_8}{I_0}h_0)$ as trial functions in the **exact** $\lambda = 2$ matrix: compression eigenvalues 1.59×10^{-12}, 3.93×10^{-7}, hence $\mu_1 \le 3.93 \times 10^{-7} \ll O(1)$ — *proven*, not measured (exact arithmetic, trial functions, min–max; the exact $\mu_1 = 2.75 \times 10^{-7}$ sits consistently below). The low spectrum of the window Weil form is a **ladder, by theorem.**

The second cell is the first excited mode — structurally. The first excited even eigenvector of the exact matrix has projection **1.0000** onto $\mathrm{span}\{\mathcal{E}h_0, \mathcal{E}h_4, \mathcal{E}h_8\}$ and only 0.414 onto the first cell $\mathrm{span}\{\mathcal{E}h_0, \mathcal{E}h_4\}$: the h_8 content is essential, and the $\mathcal{E}$-image of the mod 4 prolate tower diagonalizes the low spectrum of QW_λ to the measured precision. This is the structural heart of the trace-formula bridge, certified before being proven.

The Fuchs first-rung law, certified. On the clean window ($\lambda = 1.5$–1.8, where the FD floor 7×10^{-12} is 90× below signal): $1 - \chi_4$ tracks the CCM-quoted Fuchs asymptotic $\frac{2^{14}}{3}\sqrt{2}\pi^5 e^{-4\pi\lambda^2}\lambda^9$ at ratios $0.50, 0.56, 0.60, 0.64$ — *monotonically converging to* 1 as the asymptotic regime is entered. The rung ratio $(1-\chi_8)/(1-\chi_4)$ exceeds $c^4/50$ everywhere and scales as $c^{7.7}$ pre-asymptotically (steeper than the limiting Fuchs c^4 — more room, not less).

Theorem E.21(b) — (H3) reduced. Define the bridge inequality

$$\textbf{(B2)} \qquad \mu_1(QW_\lambda) \;\ge\; \frac{1-\chi_8(\lambda)}{\mathrm{poly}(\lambda)}.$$

Then (B2) + Fuchs (certified) + (H2) ⇒ (H3) with **four powers of c to spare**: $\mu_1/QW(\tilde{k}) \ge (1-\chi_8)/(C\,\mathrm{poly}\,(1-\chi_4)) \sim c^4/\mathrm{poly} \approx 10^5$ against the required $\lambda^{1+\delta} \approx 2.4$. Everything in (H3) except (B2) is now proven or classical; (B2) is the *precise* statement the Co-zeta trace formula must deliver, and its content is exactly what the certified cell-identification displays: the form does not dip below the second prolate defect on the orthocomplement of the ground cell.

Ledger after the strike. RH ⇐ (H1) + (H2) + (B2): simple-even (verified ×3; tower ground-cell isolation the route); the seam Rayleigh bound (measured $r/\epsilon = 0.135$; Co-zeta route); the second-rung bridge (cell identification certified at overlap 1.0000; Co-zeta route). Both remaining analytic obligations point at the **same object** — the trace formula relating QW_λ to $(P_\lambda, \widehat{P}_\lambda, \mathcal{E})$ — which is therefore, with full precision, *the* remaining mathematical frontier of this architecture. Queued: the Co-zeta/CC-2021 read-and-deploy, and the referee write-out of §§1–23. *[Historical flag — (H1) was discharged unconditionally (P∞-F §12); (H2) was decoupled from the trace formula in §25; (B2) was split in §26; the final form is* RH $\iff$ (H0′)*, §22 erratum + P∞-F §§14–17.]*

Anchors (§23). Fuchs 1964; CCM 2511.22755v1 (Fuchs quote; Figure fpro1); Courant–Fischer; Slepian school; Co-zeta (Connes 1999); CC 2021 (Weil positivity, archimedean place; arXiv 2006.13771); §§21–22; the thirteen GREEN certificates of §§11–23.

24. The trace formula read: the bridge exists, its kernel is the tower, and the two obligations are one extension

The read-and-deploy on the bridge sources (CC 2021, arXiv 2006.13771; VJ, arXiv 2106.01715) returns a sharp verdict: **the bridge object is a published theorem, its λ-instance is exactly the program's window, and its correction kernel is built from the tower defects themselves.**

What is proven (CC 2021). *Theorem 1*: for $g \in C_c^\infty([2^{-1/2}, 2^{1/2}])$ (the prime-free window $\lambda = \sqrt{2}$) with $\hat{g}$ vanishing at the poles, $W_\infty(g \star g^*) \;\geq\; \mathrm{Tr}(\vartheta(g)\,\mathbf{S}\,\vartheta(g)^*) \;\geq 0$, where $\mathbf{S}$ is **Sonin space** — the orthocomplement of $\mathrm{Range}(\mathcal{P}_1 \vee \widehat{\mathcal{P}}_1)$, the subspace seen by *neither* cutoff. Archimedean Weil positivity as the trace of a positive operator. *Theorem 3 (the trace formula, their eq. 13–14)*:

$$\mathrm{Tr}(\vartheta(f)\mathbf{S}) \;=\; W_\infty(f) \;+\; \int f(\rho)\,\epsilon(\rho)\,d^*\rho, \qquad \epsilon(\rho) \;=\; \sum_n \frac{\lambda(n)}{\sqrt{1-\lambda(n)^2}}\,\langle \xi_n | \vartheta(\rho^{-1})\zeta_n\rangle.$$

The weights $\lambda(n)/\sqrt{1-\lambda(n)^2}$ are the **cotangents of the principal angles** between the time and band cutoffs — i.e. inverse square roots of the prolate defects $1-\chi_n$. §20's angle-operator picture is confirmed *at source*: the correction kernel is tower-graded, dominated by the near-invariant cells with weight $\sim e^{+2\pi\lambda^2}$, exactly the cells our exact matrix found carrying the low spectrum (§23, overlap 1.0000). Consistency check on our data: their theorem's regime is the prime-free window, and our exact $\lambda = 1.3$ spectrum (all eigenvalues positive, ground 0.0246) certifies its conclusion independently.

What is not yet proven, stated exactly. CC 2021 establishes *positivity* (≥ 0, modulo finitely many linear conditions, error $-c|\hat{g}(0)|^2$, $c \in (13, 17)$) at $\Lambda = 1$, archimedean only. It does **not** establish: (i) the semilocal extension (primes included, window growing — the form QW_λ rather than W_∞); (ii) any *graded lower bound* on the orthocomplement of the ground cell. Hence the two outstanding obligations of §23 are, with full precision, **one task: extend CC-2021 Theorem 3 from** (Λ=1, **arch**) **to** (λ, **semilocal**) **and track the n-grading of its kernel** — (H2) is the extension evaluated on the first cell; (B2) is the same extension's second-rung coefficient on the orthocomplement. The n-dependence is already explicit in their $\epsilon(\rho)$; what must be controlled is the prime terms' interaction with the Sonin decomposition — precisely the cancellation our exact matrix *measures* happening (the $O(\lambda)$-norm prime operators reduce to $e^{-4\pi\lambda^2}$ residues on the $\mathcal{E}$-images of Fourier-invariant cells, by Poisson — §19.2's mechanism).

Ledger after the read. RH $\Leftarrow$ (H1) + [semilocal graded Sonin trace formula], where (H1) is verified $\times 3$ with a tower route, and the bracketed object is a *one-paper-shaped* extension of a published theorem whose every structural prediction our certificates have confirmed (the tower diagonalizes; the kernel is angle-graded; the prime cancellation occurs; the constants track Fuchs). This is the program's final localization of the day, and the next work is no longer reformulation: it is the extension proof itself. Queued alongside: the referee write-out of §§1–24. *[Historical flag — superseded by §25 ((H2) decoupled from the trace formula, no extension needed) and §26 ((B2) → (B2b)); (H1) discharged (P∞-F §12). The semilocal graded extension survives as the published shape of the (H0′-tail) target — P∞-F §§17.2–18.]*

Anchors (§24). CC 2021 (arXiv 2006.13771: Thm 1, Thm 3 eq. 13–14, Thm 6.11; Sonin space); VJ (arXiv 2106.01715: §2.1.2, §3; λ^2=11, $\epsilon = 2.4 \times 10^{-48}$); Co-zeta (Connes 1999); CCM 2511.22755 ("the cornerstone"); §§19–23; the thirteen GREEN certificates.

25. Theorem E.22: the radical route — (H2) decoupled from the trace formula

Attempting the semilocal extension produced something better for (H2) than an extension: a *direct mechanism*, with two short lemmas and one bookkeeping lemma, certified.

Lemma E.22.1 (radical membership) [P]**.** For h even, compactly supported, with $\int h = 0$: the *global* $\mathcal{E}(h)$ lies in the **radical of the global Weil form**: $QW(\mathcal{E}(h), g) = 0$ for all g. *Proof.* Mellin factorization: $\mathcal{M}(\mathcal{E}h)(s) = \zeta(s+\frac{1}{2}) \cdot \int h(w) w^{s-\frac{1}{2}} dw$ (valid for $\Re s > \frac{1}{2}$, extended by continuation; $\mathcal{M}_h$ entire). The zero side of $QW(\mathcal{E}h, g)$ is a sum over ξ-zeros of terms each carrying the factor $\zeta(\rho) = 0$ — *termwise zero*, unconditionally (the zero set is closed under $\rho \mapsto 1-\bar{\rho}$, so the paired evaluation also vanishes). Pole terms: $\widehat{\mathcal{E}h}(i/2) \propto \int w\,h(w)\,dw = 0$ by **evenness**; $\widehat{\mathcal{E}h}(-i/2) \propto \int h = 0$ by the **vanishing integral** — the two defining conditions of the CCM weight are exactly the two pole-killers. □ (This is the ζ-cycle mechanism (VJ eq. 1.4 territory) sharpened to radical membership; the windowed forms see $\mathcal{E}$-images as *near-null for an exact reason.*)

Lemma E.22.2 (the exact tail identity) [P]. $\mathcal{E}(h_\lambda)$ is supported in $(0, \lambda]$ (the cap $nu \le \lambda$ is exact); writing $\mathcal{E}(h_\lambda) = F_{\text{in}} + F_{\text{out}}$ (window / below-window), Lemma E.22.1 gives

$$QW_\lambda(k_\lambda) \;=\; QW(F_{\text{out}}, F_{\text{out}}) \qquad \textit{exactly}:$$

the windowed Rayleigh numerator *is* the Weil form on the below-window tail — and by the mirror (VJ 1.4), F_{out} is governed by the out-of-band content of $\widehat{h_\lambda}$, whose mass is the compression defect: $\|F_{\text{out}}\|^2 \le \text{poly}(\lambda)\,(1 - \chi_4(\lambda))$ — **linear** in the first-rung defect, not its square root.

Lemma E.22.3 (tail-form bound) [A]. $QW(F_{\text{out}}) \le C(\lambda)\|F_{\text{out}}\|^2$ with C polynomial: arch and pole terms by direct bounds; the prime sum by Cauchy–Schwarz per shift, $|(F^* * F)(n)| \le \sqrt{m(n)}\,\|F\|$ with $m(n)$ the mass of F_{out} below $1/(n\lambda)$, summable against $\Lambda(n)n^{-1/2}$ by the decay of the out-of-band Fourier tail. (The named technical point: the n-summability of the tail-mass; everything else is mechanical.) **Consequence: (H2)** $= QW_\lambda(k_\lambda) \le \text{poly}(\lambda)(1 - \chi_4(\lambda))$**, with no trace formula needed.** *[Tier upgrade — E.22.3 is now Lemma F.15 in referee form (P∞-F §16.1): arch+pole via the Suzuki screw-kernel calculus ($g''_\infty(t) = \frac{1}{2|t|} + O(1)$, a three-line log-kernel bound), primes by the per-shift Cauchy–Schwarz exactly as sketched; the constant certified flat, $C_{\text{eff}} \approx 0.2$, across five decades of defect.]*

Referee footnotes on E.22.1/E.22.2 (from the machine settlement, P∞-F §16.3). The two lemmas were stress-tested by a three-round GPD chain (M0) and **stand as stated**, with three clarifications now part of their record. *(i) Validity class:* the lemmas hold for $\mathcal{E}h$ in the **explicit-formula class** — the Mellin–Euler factorization and the termwise zero-side vanishing are statements of that class. *(ii)* $\int h = 0$ is the **finiteness** condition (it kills the $-i/2$ pole evaluation and the dual leading term in the Poisson bound). *(iii)* $h(0) = 0$ is the **Poisson-symmetry condition of the CCM weight** — *not* a radical condition: two-condition combinations ($\int h = 0$ *and* $h(0) = 0$ exactly) change nothing in the radical mechanism, and the hypothesized pole-cross corrections ($\int h \log w = \int wh = 0$) are refuted by the explicit-formula bookkeeping itself (both hypotheses retained as corpses there). The exact tail identity E.22.2 is **machine-confirmed at** 1.3% ($\lambda = 2$, all pieces above their noise floors); the residual haze at small λ is the (F64-2) far-tail float phenomenon — pole-weighted far tails of FD-prolate images — quantified and deposited as a methods rule in P∞-F §16.3.

Certificate (`empirical/p_infinity/probe_h2_linear_law.py`, GREEN): across $\lambda = 1.4$–1.7 the exact-matrix Rayleigh quotient falls by $\times 5704$ while $R/(1-\chi_4)$ spans only $\times 2.67$ (4.9, 8.0, 8.4, 13.2) — the linear law with slowly-varying constant, exactly as Lemmas E.22.1–3 predict (a $\sqrt{\ }$-law would give ratios drifting by $\times 75$).

Ledger update. RH $\Leftarrow$ **(H1)** + **(B2)** + [**E.22.3 bookkeeping**]: simple-even (verified $\times 3$); the second-rung lower bound (the *only* piece still requiring the graded Sonin/trace-formula extension of §24); and one summability lemma. (H2) has moved from "extend a published theorem" to "two proven lemmas plus bookkeeping". The frontier is now (B2) alone in trace-formula territory — and the radical mechanism suggests its shape too: on the orthocomplement of the ground cell *within the $\mathcal{E}$-image sector*, the same tail identity holds with the second cell's defect; what (B2) adds is control *off* the $\mathcal{E}$-sector, which is where Sonin space lives. That is the precise seam between what this section proves and what the §24 extension must supply. *[Historical flag — (H1) discharged (P∞-F §12); (B2) split into (B2a/b/c) in §26; the frontier evolved through P∞-F to* RH $\iff$ (H0′)*.]*

Anchors (§25). VJ (eq. 1.4; $\mathcal{E}$-map); CCM (the weight h: evenness + vanishing integral = the pole-killers); CC 2021 (Sonin complement = the off-$\mathcal{E}$ sector); Mellin–Euler factorization; §§19, 21–24; `probe_h2_linear_law.py` (the fourteenth GREEN certificate of the day).

26. Theorem E.23: the sector split — (B2) reduced to soft semilocal Sonin positivity

The radical mechanism predicts the *entire* near-null space of the window Weil form is the $\mathcal{E}$-sector. The exact matrices confirm it totally (`empirical/p_infinity/probe_b2_sector_split.py`, GREEN).

The measurement. Within the even sector, compress QW_λ onto the span of the three $\mathcal{E}$-cells ($n = 4, 8, 12$ vanishing-integral combinations) and onto its orthocomplement. At $\lambda = 2$: cell block $\{1.40 \times 10^{-12}, 2.78 \times 10^{-7}, 4.27 \times 10^{-3}\}$ — the ladder, in full; **off-$\mathcal{E}$ block: minimum** 0.568, all ten eigenvalues $O(1)$; cross-block norm 2.5×10^{-2}; *every* sub-threshold eigenvalue of the full even form is captured by the cell block ($3 = 3$; at $\lambda = 1.5$: $1 = 1$, off-$\mathcal{E}$ minimum 0.975 — and the third cell has defect $O(1)$ there, correctly *outside* the ladder: the rung count grows with λ as the tower predicts).

Theorem E.23 (the split of B2). The gap obligation decomposes: **(B2a)** *within-$\mathcal{E}$ grading* — the form on the k-th cell equals the Weil form on its below-window tail (Lemma E.22.2, exact), of size poly $\cdot (1 - \chi_{4k})$: the ladder is the tower, by the radical mechanism; what remains here is the *lower* tail-form bound, bookkeeping-class. **(B2b)** *off-$\mathcal{E}$ uniform positivity* — $QW_\lambda \geq 1/\mathrm{poly}(\lambda)$ on the orthocomplement of the $\mathcal{E}$-sector. Measured at 0.57–0.97, flat. This is the **sole genuine residue**: it is precisely CC-2021's Sonin-space positivity (their Theorem 1, $\mathrm{Tr}(\vartheta(g)\mathbf{S}\vartheta(g)^*) \geq 0$) extended semilocally — and it needs only *inverse-polynomial softness*, not exponential precision: a positivity with room, on the sector where the archimedean theorem already delivers positivity, requiring that the prime terms (which the radical mechanism shows act through exp-small residues on the $\mathcal{E}$-sector) do not destroy an $O(1)$ bound off it. **(B2c)** *mixing control* — the cross-block norm satisfies the Schur stability criterion (measured; at $\lambda = 2$, $|\mathrm{cross}| = 0.025$ against $\sqrt{\mathrm{cell} \cdot \mathrm{off}}$): perturbation-theory-class. *[Tier upgrade — now the cross-ladder law, Lemma F.16 (P∞-F §16.2): the cell-to-off-$\mathcal{E}$ coupling is graded by the tower, $\propto \sqrt{\mathrm{defect}_k}$-or-better, certified across eight decades.]*

The closing ledger of the day (§§11–26).

$$\mathrm{RH} \impliedby (\mathrm{H1}) + (\mathrm{B2b}) + \text{bookkeeping [E.22.3, B2a-lower, B2c]},$$

where (H1) is verified at three windows with a tower route to proof, and **(B2b) — soft semilocal Sonin positivity — is the single piece of genuinely new mathematics remaining in this architecture.** Its archimedean instance is a published theorem (CC 2021); its statement needs 1/poly, its measurement shows $O(1)$ flat; and its proof ground — the interaction of the prime operators with Sonin space — is exactly where the program's §16 dictionaries (passivity, gap protection, resonance widths) and the CC trace-formula machinery meet. Sixteen sections, fifteen GREEN certificates, one residue: that is the day.

The (B2b) fate (the sequel's map — added on the referee pass). The assault on (B2b) is P∞-F, and the closing ledger above is its launch point, now historical. The arc: (B2b) → F.4.D + F.5′ (the sampling-sector anatomy: the flat floor = $2\theta'(\gamma_1)$, off-$\mathcal{E}$ = the zero-representer ladder; zero-evaluation annihilation F.4.B) → F.5″ (the zero-set frame bound; the absorption identity F.7: the $\mathcal{E}$-tower is the cone's sampling debt, $K = \lambda^2 - \frac{7}{8} + O(1)$) → the merged identity (F.11: division and frame, one sampling geometry; the Hardy obstruction = RH's inner factor, F.10) → ($\mathrm{QS}_{\exp}$) + (a) (P∞-F §11) → the Suzuki weld and the **(H1) discharge** (P∞-F §12) → the withdrawn bootstrap (P∞-F §§13–14.1) → **(H0′)**: E.20's hidden hypothesis, the conserved core. Per-window, (B2b) is **discharged unconditionally** (Prop F.6: positivity by computation below the verified height); the asymptotic residue is (H0′) itself. Honest final form: $\boxed{\mathrm{RH} \iff (\mathrm{H0}')}$ through the corrected amplifier (§22 erratum), with the (H0′) anatomy in P∞-F §§17–22 — the three certified faces of ϵ_λ, the (H0′-tail) horizon localization and its measured anatomy (PNT guards the edge, the arch–prime conspiracy guards the interior, the Wiener–Hopf home), the center-out innovation chain (RH $\iff$ $d(s) > 0$ pointwise), and the remainder law $R \approx Ce^{-0.342\lambda^2}$ with its two-branch lifecycle in cumulative coordinates.

Anchors (§26). CC 2021 (Thm 1: Sonin positivity); §§21–25; Schur complement; `probe_b2_sector_split.py` (the fifteenth GREEN certificate); P∞-F (the (B2b) assault — the sequel).

Certificate index (referee audit, Draft 2)

Sixteen probes, all GREEN, deterministic, local (exact arithmetic or mpmath where load-bearing; the F64 rules apply). Paths relative to the repo root; unprefixed files live in `empirical/p_infinity/`.

- **§10.1** — `probe_e7_cross_exam_tilt_weight.py` *(repo root)* — the PNT-tilted Banach-limit weight: positive, tail, θ-homogeneous of every intermediate modulus, unconditionally — E.7's abundance demolition.
- **§10.2** — `empirical/probe_E7_ff_beurling_counterexample.py` — Theorem E.8: the fake Beurling curve ($q = 9$, $M_d \geq 0$ to $d = 300$, off-line weights 5.303/1.697) — the vN-abstract No-Go.
- **§11** — `probe_obligations_2_3_krein.py` — E.10 (B-skewness, geometric side, 10^{-13}), E.11 (FE conjugation isometry), E.12 ($\kappa(B_{\text{full}}) = 1$ exactly; zeros-only PSD witness).
- **§12** — `probe_nkappa_krein_langer.py` — E.13: Nevanlinna kernel of $-\Xi'/\Xi$ PSD on $\mathbb{C}^+$ ($\kappa = 0$ witness); counting law; grafted quadruple $\to \kappa = 2$; winding = index.
- **§13** — `probe_index_resolution_sT.py` — E.14: index = Turing discrepancy $N - N_0$; ε-graded winding; resolution-cost law $|\lambda_-| \sim d^{2.2} h^{-5.3}$.
- **§14** — `probe_krein_collision_spectral_flow.py` — E.15: Π_1 pinning (40 paths $\times$ 2001 steps, zero excursions); Honda–Tate = Krein collision at $a = 2\sqrt{q}$ exactly.
- **§15** — `probe_rank_one_determinant_spec.py` — E.16: Krein det-ratio identity; Herglotz reality; Hadamard/Hurwitz positive mode; the 4.9×10^{-9}-uniform shift migrating two zeros (weighted-spec necessity).
- **§17** — `probe_rs_gauge_body.py` — E.17: gauge identity $\Xi = -\text{env} \cdot Z$ to 10^{-31}; envelope exponent 7/4; RS skeleton; stationary-phase saddle; place-universality of $e^{-\pi|t|/4}$.
- **§18** — `probe_one_cone_two_edges.py` — E.18: the 1/8 seam deficit to 10^{-6}; three-way count match; winding neutrality vs half-field drift -0.813.
- **§19** — `probe_ccm_seam_weld.py` — E.19 engine room: exact Poisson mirror ($h(0) = 0$, $\int h = 0$); Meixner–Schäfke rates from scratch; $\hat{k}_\lambda \to \Xi$ at $\lambda^{-2.3}$; gap-transfer on 200 instances.
- **§20** — `probe_window_weil_spectrum.py` — window Weil positivity witness; near-null cluster growth $1, 3, 6, 9$; the float64 cliff-vs-ladder impasse (resolved §21).
- **§21** — `probe_ccm_exact_matrix.py` — exact QW_λ (dps 40): simple-even $\times 3$; exponentially-graded ladder; ϵ-vs-Fuchs across 13 decades; $r/\epsilon = 0.2465$.
- **§22** — `probe_quantified_weld_e2e.py` — E.20 end-to-end: min–max on real data; transfer bound at 4 strip points; $\hat{\xi}$ zeros real to 10^{-44}, $|\hat{z}_1 - \gamma_1| = 2.4 \times 10^{-9}$ at $\lambda = 2$.
- **§23** — `probe_fuchs_second_rung.py` — E.21: cliff refuted by Courant–Fischer ($\mu_1 \leq 3.93 \times 10^{-7}$ proven); second cell at overlap 1.0000; Fuchs first-rung law.
- **§25** — `probe_h2_linear_law.py` — E.22: the linear law $R/(1 - \chi_4)$ within $\times 2.67$ while R falls $\times 5704$.
- **§26** — `probe_b2_sector_split.py` — E.23: cell ladder vs off-$\mathcal{E}$ $O(1)$ block ($\min 0.568$ at $\lambda = 2$); every sub-threshold eigenvalue captured by cells.

(The §16 dictionary is underwritten by the §§11–15 probes; §24 is a source read, no new probe. The E.22.2 tail identity's direct machine confirmation at 1.3% *and the F.15/F.16 constants live in P∞-F §16's* `probe_pinfF_bookkeeping.py`*.)*